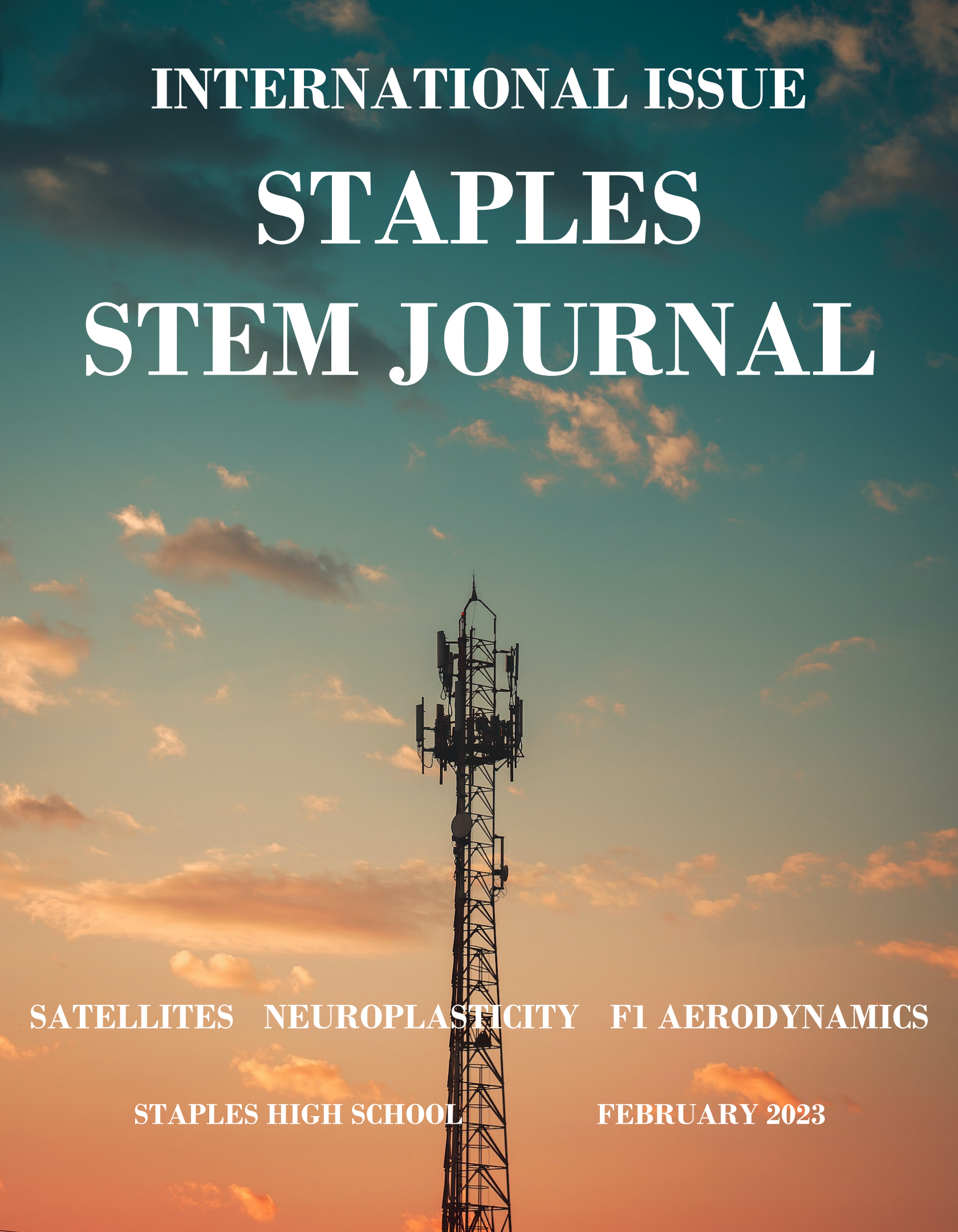
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We are thankful for the support of our excellent advisor, Ms. Amy Parent.

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The Staples STEM Journal provides an outlet for individuals to share their STEM interests with the Staples community and aims to broaden public interest and knowledge in these fields.

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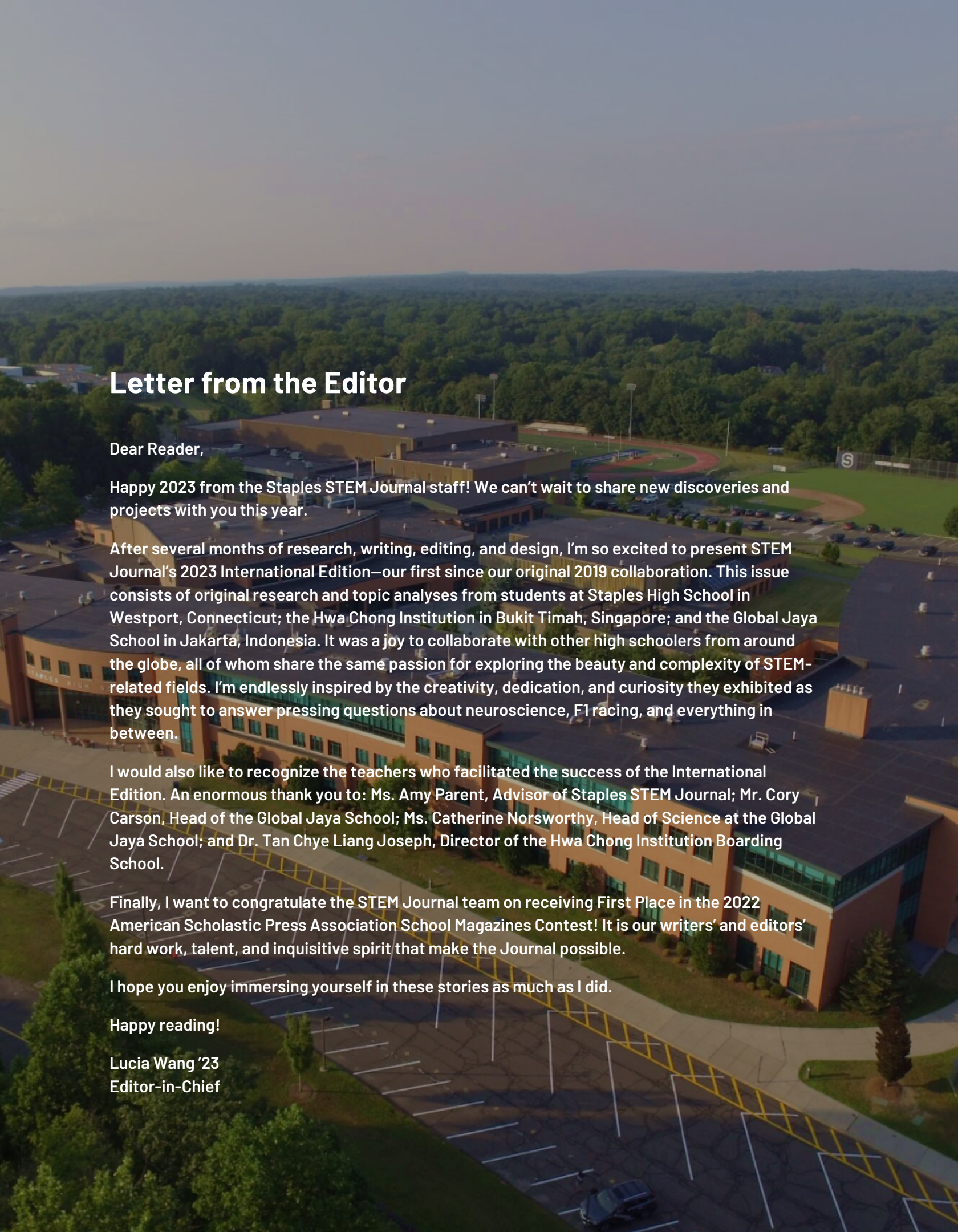
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Letter from the Editor

Dear Reader,

Happy 2023 from the Staples STEM Journal staff! We can't wait to share new discoveries and projects with you this year.

After several months of research, writing, editing, and design, I'm so excited to present STEM Journal's 2023 International Edition—our first since our original 2019 collaboration. This issue consists of original research and topic analyses from students at Staples High School in Westport, Connecticut; the Hwa Chong Institution in Bukit Timah, Singapore; and the Global Jaya School in Jakarta, Indonesia. It was a joy to collaborate with other high schoolers from around the globe, all of whom share the same passion for exploring the beauty and complexity of STEM-related fields. I'm endlessly inspired by the creativity, dedication, and curiosity they exhibited as they sought to answer pressing questions about neuroscience, F1 racing, and everything in between.

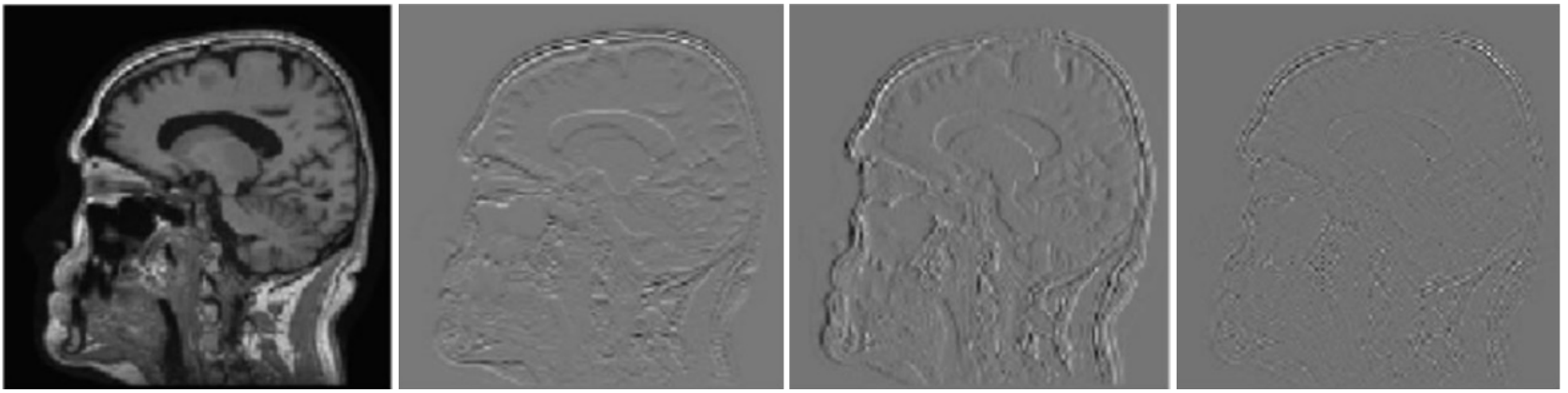
I would also like to recognize the teachers who facilitated the success of the International Edition. An enormous thank you to: Ms. Amy Parent, Advisor of Staples STEM Journal; Mr. Cory Carson, Head of the Global Jaya School; Ms. Catherine Norsworthy, Head of Science at the Global Jaya School; and Dr. Tan Chye Liang Joseph, Director of the Hwa Chong Institution Boarding School.

Finally, I want to congratulate the STEM Journal team on receiving First Place in the 2022 American Scholastic Press Association School Magazines Contest! It is our writers' and editors' hard work, talent, and inquisitive spirit that make the Journal possible.

I hope you enjoy immersing yourself in these stories as much as I did.

Happy reading!

Lucia Wang '23
Editor-in-Chief



1. Classifying Alzheimer's Disease with Machine Learning via Wavelet Transform Subband Combinations

Aalok Bhattacharya, Staples High School

Abstract

The 3-D wavelet transform has been used with machine learning techniques to help identify Alzheimer's Disease (AD) and Mild Cognitive Impairment (MCI) through magnetic resonance imaging (MRI). Although this approach resulted in high accuracy, a large number of subband permutations were obtained due to the usage of the 3-D wavelet transform, which causes redundancy and computational inefficiency during classification. To address this issue, this study discovers the combination with the minimum subbands giving a comparable accuracy with the original approach that used all subbands. The Alzheimer's Disease Neuroimaging Initiative (ADNI) was used to gain de-identified MRI scans for classification with 75% training and 25% test scans. The scans were standardized by pre-processing through skull stripping, segmentation, and smoothing using the MATLAB toolboxes Statistical Parametric Mapping 12 (SPM 12) and Computation Anatomy Toolbox 12 (CAT 12). Feature extraction was then completed through different 3-D wavelet transform subband combinations. Support vector machine (SVM) classification was used with radial basis function kernels (RBF) to screen patients for AD, MCI, or cognitively normal (CN) status 3 times per subband combination. The mean and standard deviation of the test accuracy for each combination were recorded. Among the tested combinations, the maximum mean test accuracy was 95.3%. To prevent overfitting, five-fold cross-validation was performed on the top 40% subband combinations based on the mean test accuracy. The maximum cross-validation accuracy is 97.3%. Therefore, this study shows the potential of using less 3-D wavelet transform subbands to help screen patients for AD or MCI in the future.

Introduction

Alzheimer's Disease (AD) is a progressive neurodegenerative condition that leads to brain mass loss and significantly affects the quality of life for patients and their families (National Institute on Aging, 2019). It is

classified as the most common dementia that manifests in humans (National Institute on Aging, 2019), and its prevalence in the United States of America is expected to more than double by 2050 unless any breakthrough regarding prevention or cure happens (Alzheimer's Association, 2021). Mild Cognitive Impairment (MCI) is a transitional stage in cognitive function from natural cognitive decline to AD (Henderson, n.d.). Patients with MCI are at high risk for conversion to AD and the progression of further cognitive decline (National Institute on Aging, 2019). Diagnosis of MCI and AD has primarily centered on cognitive tests administered by a physician or through brain scans, such as MRI scans (National Institute of Neurological Disorders and Stroke, 2022).

In addition to conventional methods, machine learning techniques including support vector machine (SVM), artificial neural network (ANN), deep learning (DL), and ensemble methods have shown promise identifying AD and MCI (Tanveer et al., 2020). Among them, SVM classification is one of the most excellent methods (Zhang et al., 2015) because it is able to produce high levels of classification accuracy without knowledge about the geometry and distribution of the given dataset (Cortes & Vapnik, 1995). SVM selects and optimizes a hyperplane to separate data (Bhasin & Agrawal, 2020) by maximizing the margin, or the distance from the hyperplane to the closest data points. SVM functions that separate the data (called kernels) can either be linear or non-linear (Cortes & Vapnik, 1995). However, researchers stated that SVM methods that use non-linear kernels perform well on non-linear data such as MRI scans (Zhang et al., 2015). In SVM and other machine learning classifiers, cross-validation is a procedure used to test the predictive power of a classifier on data that has not been used in the classifier before (Wang et al., 2014). In essence, it tests to make sure a classifier can generalize well and does not overfit a dataset. In k-fold cross-validation, the data set is broken into k folds, with $k-1$ folds being used as training data and then iterating the process.

While cross-validation validates SVM results, robust SVM classification itself relies on effective feature extraction. One type of feature extraction that has been used with SVMs in identifying AD and MCI is the wavelet transform, a type of transform that allows for the capture of both frequency and temporal information from a signal (Mallat, 1989). Past studies using the wavelet transform have produced promising results (Bhasin & Agrawal, 2020; Wang et al., 2014; Mishra and Deepthi, 2020; Jha et al., 2017). For example, (Bhasin & Agrawal, 2020) was able to achieve 88.34% classification accuracy for identifying MCI against cognitively normal (CN) scans. However, these studies suffered from only using a few 2-D slices of MRI scans to reduce run time and then applied the 2-D wavelet transform to those slices, which may result in missing foci of AD (Jha et al., 2017). In addition to the 2-D wavelet transform applications, applications of the 3-D wavelet transform have been performed, such as (Mishra & Deepthi, 2020), where they combined several different types of the wavelet transform such as the dual tree M-band wavelet transform or the stationary wavelet transform and then classified the data with an SVM classifier. Researchers also used the wavelet transform on 3D MRI scans followed by principal component analysis and feature fusion before final SVM classification (Ayaz et al., 2017). However, these procedures use all wavelet transform subbands, inevitably adding time to feature extraction. Subbands are created because the wavelet transform applies a low-pass (approximation coefficient creating) and high-pass (detail coefficient creating) filter in each dimension (Kai, n.d.). Figure 1 displays how subbands are created from a 3-D MRI scan. To be more specific, there are two subbands for a 1-D signal, four for a 2-D image, and eight for a 3-D image. These subbands differ in what filters have been applied in each dimension, meaning the data encoded in them is different and each subband will act different in an SVM classifier. Although the studies have demonstrated the potential of using all the subbands in AD and MCI identification, the efficacy of just using some subbands for classification is a question that needs to be answered.

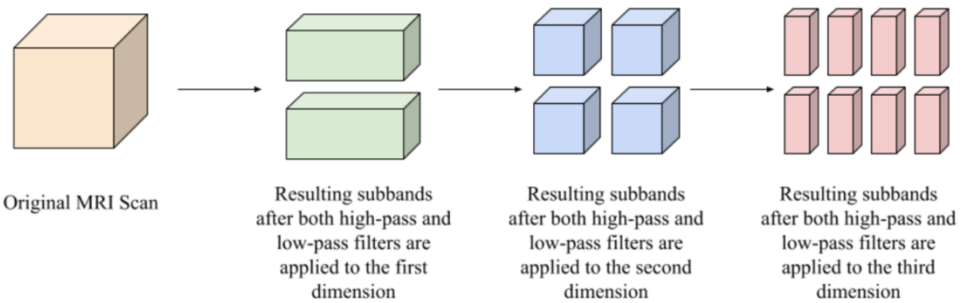


Figure 1. As high-pass and low-pass filters are applied to each dimension, a 3-D MRI scan gets divided into eight different subbands.

This study presents a machine learning solution based on selecting 3-D wavelet subband combinations of two and using a SVM method for classifying MRI scans as AD, MCI, or CN. By only using two subbands, the run time was significantly reduced while a comparable accuracy was achieved compared to past studies using all the subbands from the 3-D wavelet transform.

Methodology

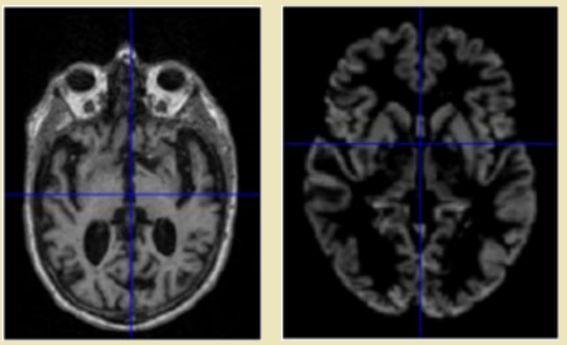
Data Acquisition

The ADNI database was used in this research, which consists of scans from MRI, positron emission tomography (PET), and other data relating to biomarkers of AD (ADNI, 2009). The objective of the database has been to test the utility of using these biomarkers to characterize the status and progression of MCI and AD (ADNI, 2009). The database has four sections: ADNI-1, ADNI-GO, ADNI-2, and ADNI-3. The ADNI-1 part of the dataset was used because it was the easiest to obtain due to it being well-established, with each 3-D scan from the section falling into three categories: AD, MCI, and CN. Only 50 scans from each category were selected due to computational limitations. As a result, this is a pilot study. The scans were selected by increasing numerical value of the file identification. 75% of these scans were used for training while 25% were used for test classification.

Each of the MRI scans were downloaded in the NIFTI format, which stores them in an accessible manner by the computer. A comma separated values (CSV) spreadsheet provided by ADNI was used to move the images into folders based on their subject condition (AD, MCI, or CN).

Image Pre-processing

Gray matter atrophy has been found to be responsible for MCI and AD progression (Apostolova et al., 2007). Pre-processing also standardizes the images. (Friston et al., 1994). Thus, pre-processing via segmentation and smoothing to capture just the gray matter of the MRI scans is necessary. The MRI scans were segmented using the Computational Anatomy Toolbox 12 (CAT 12) (Rajapakse et al., 1997) toolbox of MATLAB. These programs segmented the MRI images into grey matter, white matter, and cerebrospinal fluid. Figure 2 shows an unmodified MRI and a segmented MRI, which demonstrates the differences between them. After segmentation, the grey matter from each MRI scan was smoothed using the Statistical Parametric Mapping 12 (SPM12) (Friston et al, 1994) toolbox of MATLAB. These smoothed grey matter images were then sent for feature extraction.



A) *B)*

Figure 2. Pre-processing via SPM12 produces an MRI image without the skull and only containing gray matter: 2A) An unmodified MRI scan; and 2B) The MRI scan after segmentation.

Feature Extraction

The wavelet transform is a process by which both the frequency and time information of a signal can be captured, which applies well for non-stationary signals and images like MRIs (Wang et al., 2014). Types of wavelet transforms can be sectioned into two categories based on the orthogonality of the mother wavelet: continuous wavelet transform (CWT), which uses non-orthogonal mother wavelets, and discrete wavelet transform (DWT), which uses orthogonal mother wavelets. Since MRI scans are discrete, DWT is used in MRI classification (Bhasin & Agrawal, 2020). Equation 1 represents the wavelet expansion of a discrete function $f(x)$

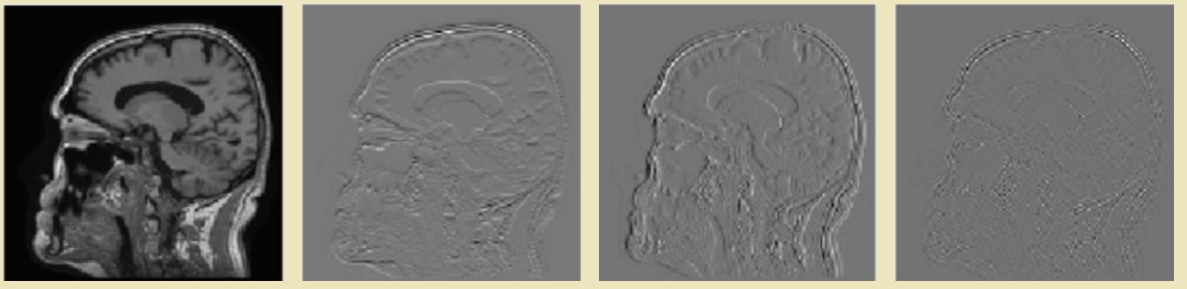
$$f(x) = \frac{1}{\sqrt{A}}\sum_{a} W_{\tau}(P,a)\tau_{P,a}(x) + \frac{1}{\sqrt{A}}\sum_{p=1}^{P}\sum_{a} W_{\psi}(p,a)\psi_{p,a}(x) \quad (1)$$

where $\frac{1}{\sqrt{A}}$ is the normalizing factor, P is the decomposition level, $\psi_{p,a}(x)$ represents the wavelet coefficients, and $\tau_{P,a}(x)$ represents scaling coefficients. $\psi_{p,a}(x)$ and $\tau_{P,a}(x)$ are discrete functions in and where $a = \{0,1,2, \dots \frac{A}{2^P} - 1\}$. Equations 2 and 3 show how the wavelet and scaling coefficients are generated (Bhasin & Agrawal, 2020):

$$W_{\psi}(p,a) = \frac{1}{\sqrt{A}}\sum_{x=0}^{A-1} f(x)\widetilde{\psi_{p,a}}(x) \quad (2)$$

$$W_{\tau}(P,a) = \frac{1}{\sqrt{A}}\sum_{x=0}^{A-1} f(x)\widetilde{\tau_{P,a}}(x) \quad (3)$$

To display how the DWT performs on an MRI scan, the 2-D DWT performed on a 2-D slice of a 3-D MRI scan. Four subbands were generated by a low-pass or high-pass filter being performed on each of the two dimensions in 2-D image. Each subband is represented as a two-letter key, representing what subbands are in which dimension. An "a" represents an approximation subband produced by a low-pass filter and a "d" represents a detail subband produced by a high-pass filter. Figure 2 shows the four resulting subbands after this procedure.



A) B) C) D)

Figure 2. The visualization of the 2-D DWT performed on a model slice with all four subband patterns: 2A: Approximation subbands in both dimensions (aa); 2B: approximation subbands in the horizontal dimension and detail subbands in the vertical dimension (ad); 2C: Detail subbands in the horizontal direction and approximation coefficients in the vertical direction (da); 2D: Detail subbands in both dimensions (dd).

However, the application of the DWT on 2-D slices of MRI scans is prone to missing foci of AD pathology that are necessary for accurate classification (Jha et al., 2017).

The application of the 3-D DWT was the primary focus of this study for its use in feature extraction of the entire 3-D MRI while only using certain subband combinations. After pre-processing, each scan was taken and resized into a standardized size. From there, the 3-D DWT was executed using the biorthogonal 1.3 wavelet with the PyWavelets package (access the PyWavelets documentation at https://pywavelets.readthedocs.io/en/latest/) of Python 3.9.6. Different subband combinations were selected and stored for later classification. To get multiple information points but limit the amount of computational capacity needed to calculate the classification, each subband combination consisted of two of the eight possible subbands at the 3-D level, resulting in a total of 36 subband combinations.

Classification and Cross-Validation

The classification of the final features in this experiment was done by a SVM classifier. Each subband combination was classified three separate times with the SVM. The C value, or regularization parameter, was ten in all trials, and the tolerance was 1×10^{-3}. Since linear SVMs do not perform well on non-linear data (Zhang et al., 2015), non-linear radial basis function (RBF) kernels were used. The test classification accuracy was recorded each time the subband combination was run through the SVM. The mean and standard deviation (SD) of the test accuracies for each subband combination were then calculated.

The top 40% of the subband combinations by mean test accuracy were selected for cross-validation. Five-fold cross-validation was used to validate the performance of these combinations.

Results and Discussion

After the 3-D DWT was performed on 150 MRI scans, the feature extracted scans were run through a SVM classifier with RBF kernels and C-value of 10 three times per subband combination. The test classification accuracy was measured for each of the three runs, and then the mean and SD of the test classification accuracies were calculated. Each subband is represented by a three-letter key detailing whether there is an approximation subband (produced by a low-pass filter) or a detail subband (produced by a high-pass filter) in the x, y, and z dimensions. An "a" represents an approximation subband and a "d" represents a detail subband. Table 1 shows the mean and SD of the test accuracy for three runs per subband combination. The minimum mean test accuracy was 75.2% and the maximum mean test accuracy was 95.7%. The top 40% of subband combinations sorted by mean test accuracy were selected for five-fold cross-validation. Table 2 shows the mean and SD of cross-validation accuracy per subband combination, which are sorted by highest mean cross-validation accuracy and lowest SD. Five-fold cross-validation of the selected subband combinations yielded a minimum mean accuracy of 91.3% and a maximum of 97.3%.

Table 1. The mean and SD of the test accuracy for three runs each, shown as (mean ± SD). The bolded accuracies were selected for cross-validation.

aaa	0.855 ± 0.030							
aad	0.872 ± 0.032	0.889 ± 0.030						
ada	0.863 ± 0.015	**0.934 ± 0.061**	**0.923 ± 0.044**					
add	0.752 ± 0.074	**0.940 ± 0.030**	**0.923 ± 0.026**	0.897 ± 0.051				
daa	0.838 ± 0.030	0.906 ± 0.059	0.897 ± 0.068	0.897 ± 0.051	0.855 ± 0.116			
dad	0.812 ± 0.015	**0.940 ± 0.015**	**0.940 ± 0.059**	0.889 ± 0.053	0.863 ± 0.097	**0.940 ± 0.030**		
dda	0.803 ± 0.039	**0.949 ± 0.068**	**0.949 ± 0.026**	**0.949 ± 0.051**	0.915 ± 0.053	0.915 ± 0.015	**0.923 ± 0.027**	
ddd	0.863 ± 0.090	**0.957 ± 0.053**	**0.923 ± 0.077**	0.915 ± 0.126	**0.923 ± 0.051**	0.906 ± 0.065	0.897 ± 0.077	0.880 ± 0.074
	aaa	**aad**	**ada**	**add**	**daa**	**dad**	**dda**	**ddd**

Table 2. The mean and SD of cross-validation accuracy for five-fold cross-validation, shown as (mean ± SD)

Subband Combination	Mean and SD of Cross-Validation Accuracy
aad, add	0.973 ± 0.013
dda, dda	$0.967 \pm 1.1 \times 10^{-6}$
ada, add	0.960 ± 0.033
ada, dad	0.960 ± 0.033
aad, ada	0.953 ± 0.045
ada, dda	0.953 ± 0.050
ada, ddd	0.953 ± 0.050
aad, dad	0.947 ± 0.039
add, dda	0.947 ± 0.054
daa, ddd	0.940 ± 0.053
aad, dda	0.920 ± 0.068
ada, ada	0.920 ± 0.068
dad, dad	0.913 ± 0.016
aad, ddd	0.913 ± 0.039

This study found that the selection of only two subband patterns can still produce accurate classification. The maximum cross-validation classification accuracy was 97.3%, compared to the 76.0% accuracy found by Ayaz et al. or the 92.8% accuracy found by Sujatha Kumari et al. (Sujatha Kumari et al., 2021). This means that despite using less subbands and needing less computational power, this study's procedure was able to produce better classification results than previous proposed methods using the 3-D DWT. These results show that the use of only a few subbands in specific subband combinations may be a viable innovation to increase the efficiency of screening patients for AD or MCI via MRI.

Further research is required to see if the method used in this study is applicable to larger data sets. In addition, it is possible the accuracy can be further increased by using a convolutional neural network (CNN) as the classifier.

Conclusion

This study finds that the use of only certain subband combinations may be a viable solution to creating a more efficient method of 3-D DWT on MRI scans to classify patient cognitive status as AD, MCI, or CN. The maximum mean test classification accuracy was 95.7% from the "aad, ddd" subband combination, and the maximum five-fold cross-validation accuracy was 97.3% from the "aad, add" subband combination. Despite requiring less computation capacity and less steps, these results were better than previous proposed procedures to screen patients for AD or MCI based on performing the 3D-DWT on MRI scans. This result shows that using a limited 3D-DWT subband method may be able to increase the efficiency of classifying patients for AD or MCI with MRI. However, this procedure needs to be validated on larger sample sizes of MRI scans.

Acknowledgements

I would like to thank Xiangyi Cheng for her mentorship and guidance of me through this project.

References

ADNI | About. (2009). Usc.edu. http://adni.loni.usc.edu/about/

Ayaz, A., Ahmad, M. Z., Khurshid, K., & Kamboh, A. M. (2017). MRI based automated diagnosis of Alzheimer's: Fusing 3D wavelet-features with clinical data. 2017 39th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC). https://doi.org/10.1109/embc.2017.8037048

Apostolova, L. G., Steiner, C. A., Akopyan, G. G., Dutton, R. A., Hayashi, K. M., Toga, A. W., Cummings, J. L., & Thompson, P. M. (2007). Three-Dimensional Gray Matter Atrophy Mapping in Mild Cognitive Impairment and Mild Alzheimer Disease. Archives of Neurology, 64(10), 1489. https://doi.org/10.1001/archneur.64.10.1489

Bhasin, H., & Agrawal, R. K. (2020). A combination of 3-D discrete wavelet transform and 3-D local binary pattern for classification of mild cognitive impairment. BMC Medical Informatics and Decision Making, 20(1). https://doi.org/10.1186/s12911-020-1055-x

Cortes, C., & Vapnik, V. (1995). Support-vector networks. Machine Learning, 20(3), 273–297. https://doi.org/10.1007/bf00994018

Facts and Figures. (2021). Alzheimer's Disease and Dementia. https://www.alz.org/alzheimers-dementia/facts-figures#prevalence

Friston, K. J., Holmes, A. P., Worsley, K. J., Poline, J.-P. ., Frith, C. D., & Frackowiak, R. S. J. (1994). Statistical parametric maps in functional imaging: A general linear approach. Human Brain Mapping, 2(4), 189–210. https://doi.org/10.1002/hbm.460020402

Henderson, V. (n.d.). Mild Cognitive Impairment. Retrieved February 15, 2022, from https://med.stanford.edu/content/dam/sm/adrc/documents/adrc-information-sheet-mild-cognitive-impairment.pdf

Jha, D., Kim, J.-I., & Kwon, G.-R. (2017). Diagnosis of Alzheimer's Disease Using Dual-Tree Complex Wavelet Transform, PCA, and Feed-Forward Neural Network. Journal of Healthcare Engineering, 2017, 1–13. https://doi.org/10.1155/2017/9060124

Kai, S., Li, K., & Selesnick, I. (n.d.). Wavelet Software at Brooklyn Poly. Eeweb.engineering.nyu.edu. Retrieved February 8, 2022, from https://eeweb.engineering.nyu.edu/iselesni/WaveletSoftware/standard3D.html

Leifer, B. P. (2003). Early diagnosis of Alzheimer's disease: clinical and economic benefits. Journal of the American Geriatrics Society, 51(5 Suppl Dementia), S281-288. https://doi.org/10.1046/j.1532-5415.5153.x

Mallat, S. G. (1989). A theory for multiresolution signal decomposition: the wavelet representation. IEEE Transactions on Pattern Analysis and Machine Intelligence, 11(7), 674–693. https://doi.org/10.1109/34.192463

Mishra, S. K., & Deepthi, V. H. (2020). Brain image classification by the combination of different wavelet transforms and support vector machine classification. Journal of Ambient Intelligence and Humanized Computing, 12(6), 6741–6749. https://doi.org/10.1007/s12652-020-02299-y

National Institute on Aging. (2019, May 22). Alzheimer's Disease Fact Sheet. National Institute on Aging. https://www.nia.nih.gov/health/alzheimers-disease-fact-sheet

Rajapakse, J. C., Giedd, J. N., & Rapoport, J. L. (1997). Statistical approach to segmentation of single-channel cerebral MR images. IEEE Transactions on Medical Imaging, 16(2), 176–186. https://doi.org/10.1109/42.563663

Sujatha Kumari, B. A., Yadiyala, A. G. V., Aruna, B. J., Radha, C., & Shwetha, B. (2021). Early Detection of Mild Cognitive Impairment Using 3D Wavelet Transform. Data Intelligence and Cognitive Informatics, 445–455. https://doi.org/10.1007/978-981-15-8530-2_36

Tanveer, M., Richhariya, B., Khan, R. U., Rashid, A. H., Khanna, P., Prasad, M., & Lin, C. T. (2020). Machine Learning Techniques for the Diagnosis of Alzheimer's Disease. ACM Transactions on Multimedia Computing, Communications, and Applications, 16(1s), 1–35. https://doi.org/10.1145/3344998

The Dementias: Hope Through Research | National Institute of Neurological Disorders and Stroke. (n.d.). Www.ninds.nih.gov. Retrieved January 29, 2022, from https://www.ninds.nih.gov/Disorders/Patient-Caregiver-Education/Hope-Through-Research/Dementia-Hope-Through-Research#diagnosis

Wang, X., Nan, B., Zhu, J., & Koeppe, R. (2014). Regularized 3D functional regression for brain image data via Haar wavelets. The Annals of Applied Statistics, 8(2). https://doi.org/10.1214/14-aoas736

Ye, D. H., Pohl, K. M., & Davatzikos, C. (2011). Semi-supervised Pattern Classification: Application to Structural MRI of Alzheimer's Disease. 2011 International Workshop on Pattern Recognition in NeuroImaging, 1–4. https://doi.org/10.1109/prni.2011.12

Zhang, Y., Dong, Z., Phillips, P., Wang, S., Ji, G., Yang, J., & Yuan, T.-F. (2015). Detection of subjects and brain regions related to Alzheimer's disease using 3D MRI scans based on eigenbrain and machine learning. Frontiers in Computational Neuroscience, 9. https://doi.org/10.3389/fncom.2015.00066



2. Essential oils from thyme and rosemary in combination as an antibiotic-sparing agent to treat *Escherichia coli*-caused urinary tract infections

Mallika Subramanian, Staples High School

Abstract

Antimicrobial resistance (AMR) is one of the largest healthcare emergencies today, with urinary tract infections (UTIs) caused by antibiotic-resistant uropathogenic *Escherichia coli* (*E.coli*) (UPEC) as a critical offender. The development of biofilms, matrices of UPEC, further complicate treatment. Antibiotics struggle to pierce the biofilm, leading to longer infections and antibiotic use, increasing AMR. Little has been done to identify efficient UTI treatments which inhibit both bacterial and biofilm growth. Previous research demonstrates that essential oils (EOs), especially *T.zygis* (thyme) and *R.officinalis* (rosemary) EOs, can combat bacterial growth as effectively as antibiotics, as well as inhibit biofilm growth. EO combinations have shown enhanced antibacterial and antibiofilm activity over individual EOs, but ratios of EOs in combination have not been optimized. *T.zygis* and *R.officinalis* EOs were tested on *E.coli* in different ratios to establish an optimal EO combination to inhibit bacterial and biofilm activity. Agar disk diffusion evaluated antibacterial activity(n=2) and a colony forming unit/mL assay measured antibiofilm activity(n=30). 100% *T.zygis*, 0% *R.officinalis* and 90% *T.zygis*, 10% *R.officinalis* had the highest antibacterial activities, with similar activity to ciprofloxacin control. 90% *T.zygis*, 10% *R.officinalis* and 60% *T.zygis*, 40% *R.officinalis* had the highest antibiofilm activities, with inhibition levels of 80.89% and 80.09%, respectively. 90% *T.zygis*, 10% *R.officinalis* was more effective than *T.zygis* (70.87% inhibition;p=0.03034) or *R.officinalis* (37.95% inhibition;p=0.0011) alone, and the most effective treatment overall. These results could indicate EO combinations for utilization in alternative antibiotic-sparing treatments for UPEC-caused UTIs.

1. Introduction

Antimicrobial resistance (AMR) is one of the biggest healthcare emergencies today (WHO, 2021), with *E. coli* resistance against antibiotics being one of the most urgent issues, due to its ability to cause a plethora of diseases (Maduppa, 2019). Urinary tract infections (UTIs) are one of the most prevalent diseases, affecting at least 274 million people annually worldwide (GBD, 2018), and having an incidence of 50-60% in the lifetime of adult women (Medina & Castillo-Pino, 2019). Uropathogenic *E. coli* (UPEC) causes 75% of uncomplicated UTIs and 65% of complicated UTIs (Flores-Mireles et al., 2015). Due to the abuse of antibiotics for this common

disease, UPEC has become resistant to many kinds of antibiotics (Table 1), specifically fluoroquinolones and β-lactams, such as ciprofloxacin and amoxicillin, respectively (Flores-Mireles et al., 2015; Wu et al., 2021; Smieszek & Pouwels, 2018). Fluoroquinolones and β-lactams are used to treat many other severe infections, and resistance to them can create problems concerning lack of treatment for severe disease (WHO, 2021). The rate of AMR is rising, and because of that, the rate of multi-drug resistant bacteria is also rising (WHO, 2021). This kind of resistant bacteria is a threat to anyone who is susceptible to get a UTI caused by UPEC (MacKinnon et al., 2020).

Table 1. Description of Common Antibiotics that Treat E. coli *Infections, their Antibiotic Class, and Percent* E. coli *Resistance*

Antibiotic (Madappa & Stuart, 2019)	**Antibiotic Class**	***E. coli* Resistance (%)**	**References**
Amoxicillin	β-lactam	72%	(Nji et al., 2021)
Ciprofloxacin	Fluoroquinolones	8%-65%	(*High Levels of Antibiotic Resistance Found Worldwide, New Data Shows*, 2018)
Trimethoprim/sulfamethoxazole (Co-trimoxazole)	Sulfonamide	63%	(*Co-Trimoxazole*) (Nji et al., 2021)
Ampicillin/sulbactam	β-lactam	> 50%	(Bryce et al., 2016)
Levofloxacin	Quinolone	38.3%	(Terahara & Nishiura, 2019)
Doxycycline	Tetracycline	37.5%	(*Doxycycline (Oral Route) Description and Brand Names*) (Dimitrova et al., 2021)
Aztreonam	Monobactam	1.4-9.4%	(*Aztreonam Injection*) (Cho et al., 2011)
Nitrofurantoin	Nitrofuran	0.8-1.7%	(*Nitrofurantoin - Side Effects, Uses, Dosage, Overdose, Pregnancy, Alcohol*, 2015) (Bryce et al., 2016)
Meropenem	Carbapenem	0.2%	(*Merrem IV (Meropenem): Uses, Dosage, Side Effects, Interactions, Warning*) (Nordmann & Poirel, 2019)

Complication of Current Bacterial Infection Treatments by Biofilms

AMR is further complicated by the creation of biofilms by bacteria. Biofilms are the assemblage of microbial cells that are enclosed in a matrix of polysaccharide material, and are responsible for more than 80% of all microbial infections (Römling & Balsalobre, 2012). Biofilm development is crucial for the persistence of UPEC in the urinary tract and its ability to cause UTIs (Sanchez et al., 2013). Bacterial biofilms become resistant to antibiotics because the antibiotics struggle to pierce the polysaccharide matrix (Costerton et al., 1999). This results in recurring biofilm infections, which bring about cycles of antibiotics, and increased resistance to antibiotics (Costerton et al., 1995), effectively rendering antibiotics useless against biofilms.

Essential Oils as an Alternative Therapy

Essential oils offer promising evidence as alternative treatments against both bacterial and biofilm growth. Although there have been many studies proving the effectiveness of EOs in monotherapy against bacterial growth (Lagha et al., 2019; Mahizan et al., 2019; Williams, 2018) and biofilm production of *E. coli* (Cáceres et al., 2020, Kerekes et al., 2019, Millezi et al., 2019, Neyret et al., 2014), fewer studies have investigated the potential of EOs in combination.

In an experimental procedure on the antibacterial effects of EO monotherapy and combination therapy, Williams, 2018 demonstrated that pine, orange, oregano, and coriander EOs have higher antibacterial effects in combination compared to their individual use. In addition, it showed that specific ratios of the combinations had greater antibacterial efficacy than other ratios. Bassolé & Juliani demonstrated that EOs and EO components can have synergistic, additive, and antagonistic effects on UPEC. The results of this study demonstrated that phenolic and monoterpenoid terpene (EO components) combinations are actively synergistic against *E. coli*. *R. officinalis* EO was also shown to have synergistic interactions with other EOs.

EOs have also been shown to work against biofilm growth when used in combination with other drugs. Demonstrated in Algburi et al.,, various antibiotics were tested along with natural components to measure the interaction between them when acting against biofilm growth. Thyme oil and *Pelargonium graveolens* (rose geranium) were tested with ampicillin, penicillin, cloxacillin, cephalothin, methicillin, novobiocin, vancomycin, and norfloxacin, and showed synergistic effects, as well as enhancing effects of the antibiotics. Terpinen-4-ol, an EO component, was also tested with ciprofloxacin and demonstrated synergistic effects. Neyret et al., 2014 reported that thymol and carvacrol (both phenolic components) in combination have higher antibiofilm activity in combination compared to their respective individual antibiofilm activities.

Thymus zygis (T. zygis) and *Rosmarinus officinalis (R. officinalis)* are both medicinal plants derived from the common name thyme and rosemary, respectively. Their respective essential oils(EOs) have been used for medicinal purposes (Bukovská et al., 2007). Demonstrated in Lagha et al., 2019, both *T. zygis* EO and *R. officinalis* EO proved to have antibacterial and antibiofilm capabilities when tested individually on UPEC. However, *T. zygis* EO had the highest antibacterial activity out of all the EOs, having strong inhibitory action against 90% of the *E. coli* tested, and *R. officinalis* EO had the highest antibiofilm activity, inhibiting up to 94.75% of biofilms. Both EOs are also mainly composed of monoterpenoid terpenes, with the main compounds in *T. zygis* EO being linalool and terpinen-4-ol, and the main compound in *R. officinalis* EO being 1,8-cineole (Lagha et al., 2019). The terpenes have been attributed to the antibacterial activity of the two EOs (Lagha et al., 2019), and have also been shown in other research to be effective against *E. coli* (Mahizan et al., 2019).

The benefits of *T. zygis* and *R. officinalis* EOs and their components against *E. coli* have been proven (Lagha et al., 2019)(Bassolé & Juliani, 2012), but the effectiveness of them in combination for optimization of both antibiofilm activity and antibacterial activity remains to be established.

New solutions need to be studied to treat UTIs caused by UPEC in order to decrease AMR levels, which could be done by studying treatments that address both bacterial growth and biofilm production (Høiby et al., 2015). The aim of this study was to investigate and ultimately demonstrate the heightened effectiveness of *T. zygis* and *R. officinalis* EOs in combination by effectively optimizing activity against biofilm production and bacterial growth of UPEC. It was hypothesized that the two EOs would work synergistically against UPEC when combined because they are both mainly made of monoterpenoid terpenes. It was also predicted that the EO treatments of 40% *T. zygis*, 60% *R. officinalis*, 50% *T. zygis*, 50% *R. officinalis*, or 60% *T. zygis*, 40% *R.*

officinalis (Groups 7, 8, and 9 in Table 2) would be the most effective due to previous research on specific ratios (Williams, 2018).

2. Materials and Methodology

2.1. Population

The model was a strain of non-pathogenic *E. coli* called *E. coli* K-12 that was obtained from Sacred Heart University.

2.2. Materials

Essential Oils

T. zygis and *R. officinalis* EOs were purchased from Eden Botanicals. Essential oils treatments were created to be 1000 µL total each, and formulated to have different percentages of each essential oil(shown in Table 2) using a calibrated pipette to combine oils in centrifuge tubes. To dilute both essential oils to 2500 mg/mL before creating treatments, 675 µL of *T. zygis* EO was put in 135 µL of fractionated coconut oil in a centrifuge tube and 675 µL of *R. officinalis* EO was also put in 135 mL of fractionated coconut oil in a centrifuge tube. These solutions were made as many times as necessary when they were all used up. Both essential oils were stored at approximately 21°C in the tubes until needed for use (Doterra, n.d.).

Antibiotics

For the antibiotic testing in the agar disk diffusion procedure, ciprofloxacin and penicillin antibiotic disks were purchased and used.

Table 2. Percentage by Volume of EO/antibiotic in each treatment.

	Group												
Treatment	1	2	3	4	5	6	7	8	9	10	11	12	13
T. zygis EO			0%	10%	20%	30%	40%	50%	60%	70%	80%	90%	100%
R. officinalis EO			100%	90%	80%	70%	60%	50%	40%	30%	20%	10%	0%
Ciprofloxacin (+ control)	✓												
Penicillin (- control)		✓											

2.3. Study Design

This was an *in vitro* experimental study consisting of 2 main procedures: the Kirby-Bauer agar disk diffusion method for antimicrobial susceptibility/antibacterial activity and a CFU/mL assay for biofilm inhibition.

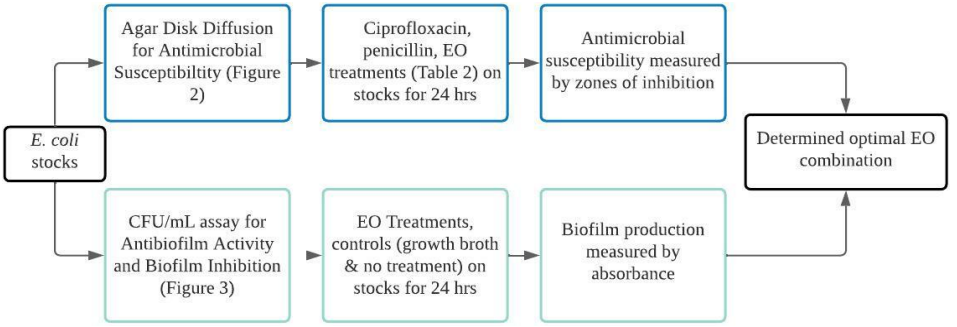


Figure 1. Overview of Study Design - Agar Disk Diffusion to Test for Antimicrobial Susceptibility and CFU/mL Assay to Test for Biofilm Inhibition.

2.4. Procedures

2.4.1. Antimicrobial Susceptibility Testing

Antimicrobial susceptibility was examined through the Kirby-Bauer agar disk-diffusion method (Balouiri et al., 2016). The entire procedure was carried out on a bench. *E. coli* stocks were grown at 37°C for 18-24 hrs in MHA broth before being spread on MHA plates. MHA plates were inoculated using a sterile swab to streak the bacteria on. 4 6mm filter paper disks were impregnated with treatment solutions (20µL/disk) and then placed on the surface of the MHA plates. The plates were then incubated at 37°C for 24 hours. Antibacterial activity was quantified by measuring the inhibition zone diameters (mm) (Figure 2) around the disks using a ruler.

For the cipro and penicillin antibiotic discs, *E. coli* was spread on the antibiotic plates, and the discs were put directly on the plates. The rest of the procedure carried on as described above.

The experiment was duplicated, and the mean zone diameter was recorded for each group (Lagha et al., 2019).

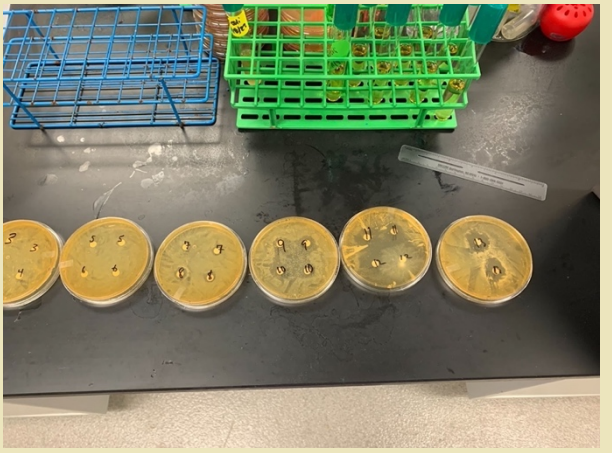


Figure 2. Zones of inhibition of groups 3-13.

2.4.2. Biofilm Formation and Inhibition

Biofilm formation by *E. coli* was observed using the crystal violet CFU/mL assay method (see figure 3 for procedure) (Wilson, 2017). This entire procedure was also carried out on a bench. The *E. coli* cultures were grown at 37°C for 18-24 hrs in Trypticase Soy broth (TSB). The suspended bacteria was then transferred to a 96-well microtiter plate using a pipette, with 100 µL in each well. 100 µL of EO solutions emulsified in TSB

(see *2.4.2.1.*) were then added to their respective wells. The total volume of each well was 200 µL (Lagha et al., 2019). The controls were pure TSB and no treatment (see Figure 3 for well distributions). The plates were incubated at 37°C for 24 hrs, and the crystal violet procedure (described below) was carried out.

2.4.2.1. Essential Oil Emulsification in TSB Supplement

The EO emulsification in TSB supplement occured in a 1:1 ratio by mixing 10 mL of EO treatment with 10 mL of TSB supplement and stirring gently. This emulsification happened for each EO treatment.

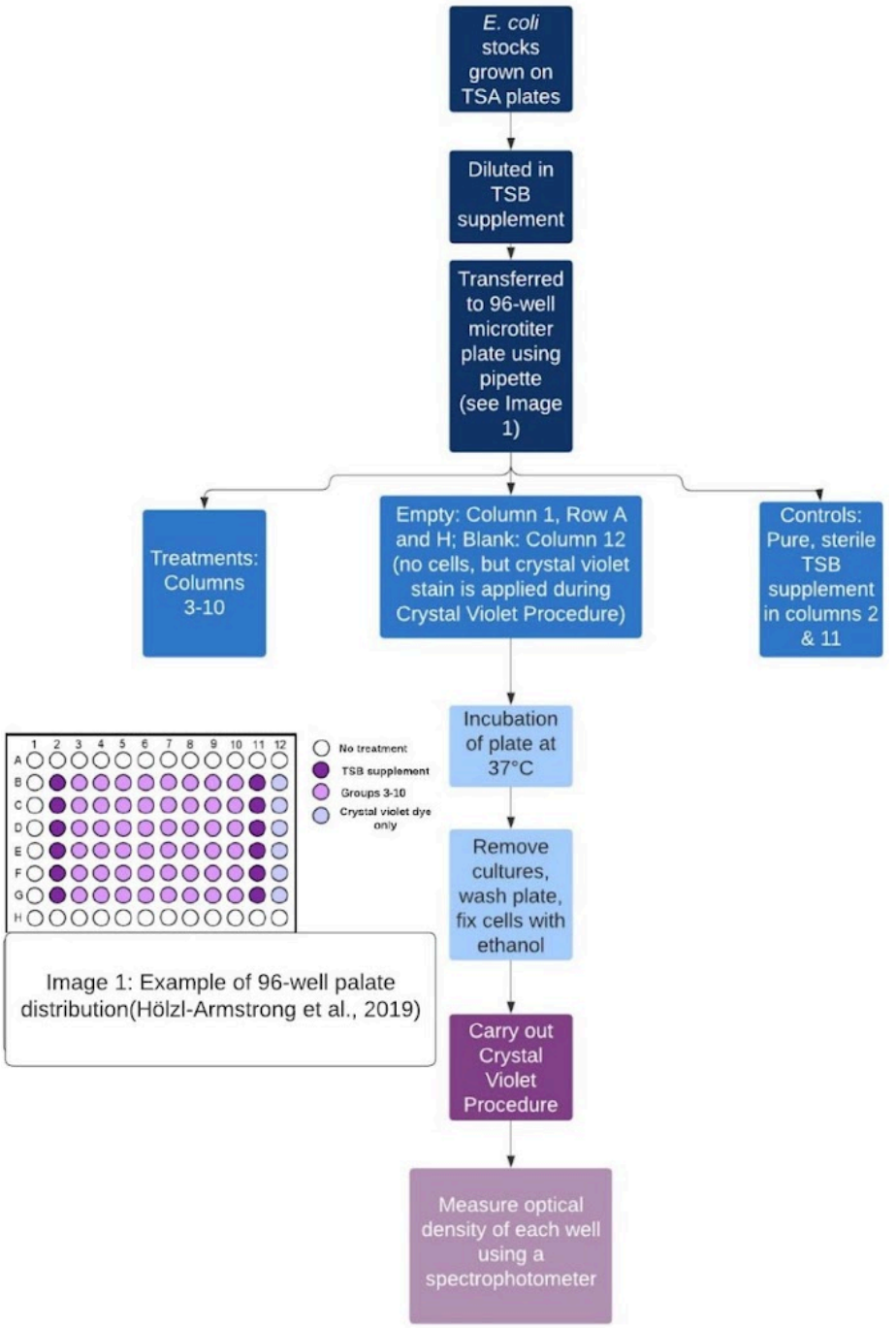


Figure 3. CFU/mL assay procedure for biofilm inhibition, including distribution of 96-well plate.

2.4.2.1. Crystal Violet Procedure

Culture mediums were carefully removed from the wells of the 96-well microtiter plate using a pipette. 0.2 mL of Phosphate Buffer Solution (PBS) was warmed up to 20℃, and then was used to wash the plate. Each well was washed with 150µL of PBS. The PBS was then removed and 50 µL of crystal violet solution was added to each well. The plate was then incubated at 20℃. After, the plate was immersed in a large beaker filled with tap water in order to wash. This process was repeated in duplicate, with the tap water being changed between washes. The plate was drained upside down on paper towels. 100 µL of 1% sodium dodecyl sulfate (SDS) was applied to each well in order to solubilize the stain. The plate then went into an orbital shaker incubator until the color of the wells was uniform (*Crystal Violet Assay*, n.d.; Hölzl-Armstrong et al., 2019).

2.4.2.2. Biofilm Production Quantification

Quantification of biofilm production started with measuring the optical density of each well of the 96-well microtiter plate. After the plate went through the crystal violet procedure, it went into a 96-well spectrophotometer at 570 nm. A darker shade of violet would display a higher absorption, and therefore, more biofilm (Figure 4).

Biofilm inhibition % was calculated using the equation below, as described in Lagha et al.

$$\% \text{ Inhibition} = 100 - [(\text{absorbance of sample}/\text{absorbance of control}) * 100]$$

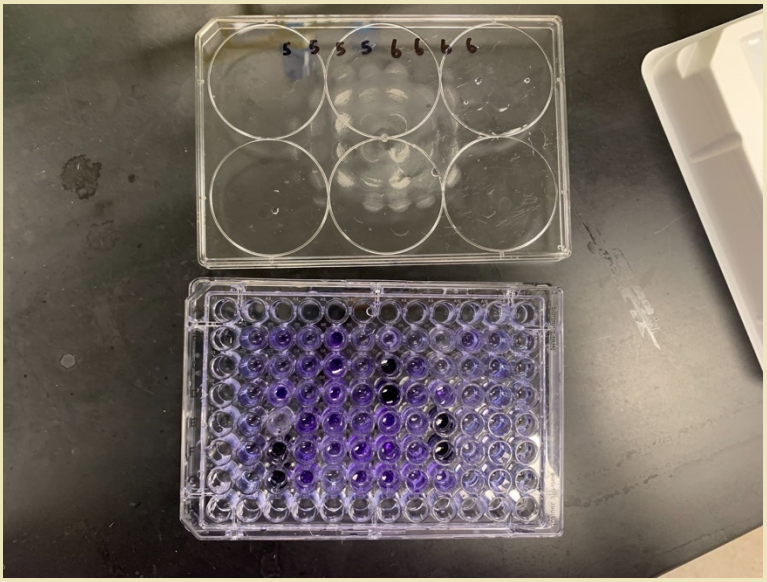


Figure 4. Different amounts of biofilm depicted by different shades of violet after crystal violet assay on groups 5 and 6.

Statistical Analysis

Statistical analysis was conducted using 2-sample t-tests. Results with p-values < 0.05 were considered to be statistically significant.

For the disk diffusion assay data, the treatments were all compared to the positive control, which was ciprofloxacin.

For the crystal violet CFU/mL assay, the biofilm inhibition % of treatments were all compared to group 13 because that treatment was just the *R. officinalis* EO, which had been proven to have high biofilm inhibition levels, even higher than *T. zygis*. By comparing to group 13, it was identified whether or not the combination would be more effective than an individual oil or not.

2.5. Limitations of Study

This experiment was aimed to research alternative medicine for UTIs caused by UPEC, however, due to lab availability and safety protocols, UPEC was not viable to use as an experimental model. However, non-pathogenic *E. coli* was acceptable to use in place of UPEC. This is because the antibacterial mechanism of action of EOs is to diffuse through the cytoplasmic membrane of the bacteria (Saad et al., 2013), and because non-pathogenic *E. coli* has a cytoplasmic membrane (Silhayy, 2015). Outer-membrane vesicles (OMVs) are also used by UPEC to cause infection (Terlizzi et al., 2017) which is recognized by the body's immune responses, and non-pathogenic *E. coli* also release OMVs that cause immune responses in the body (Behrouzi et al., 2018), so the anatomy of non-pathogenic *E. coli* is similar to the anatomy of UPEC. This means that the results of this study can be applied to UPEC when discussing alternative medicine for UTIs. In addition, the use of non-pathogenic bacteria removes any ethical or safety concerns associated with using UPEC.

3. Results

3.1. Antibacterial Activity of Essential Oil Combinations

Table 3. Disk diffusion assay results.

	Treatment												
	1	2	3	4	5	6	7	8	9	10	11	12	13
Avg zone of inhibition (mm)	24.5	0	2.75	4	8.75	5.75	11.75	14.75	18.5	17.5	20.5	28.67	27.5
Standard Deviation	3.1091	0	0.25	1	0.75	0.25	1.75	0.25	0.5	1.5	0.5	2.31	2.5
P-value compared to group 1	N/A	N/A	7.05 E-4	3.32 E-4	0.001	0.001	0.004	0.008	0.028	0.021	0.042	0.049	0.304

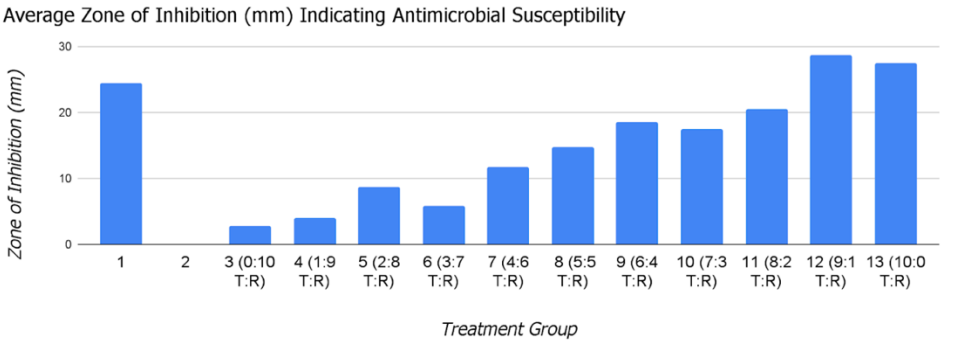


Figure 5. Disk diffusion assay portrays effectiveness of inhibiting bacterial growth.

Out of the 11 essential oil combinations, groups 12 (90% *T. zygis*, 10% *R. officinalis*) and 13 (100% *T. zygis*, 0% *R. officinalis*) had the highest antibacterial activities with average zones of inhibition of 28.67 mm and 27.5 mm, respectively. A 2-sample t-test comparing groups 12 (p=0.0487)and 13 (p=0.304) with group 1 (ciprofloxacin control) demonstrated that they both had similar antibacterial activities to the control. Group 3 (0% *T. zygis*, 100% *R. officinalis*) had the lowest antibacterial activity with an average zone of inhibition of 2.75 mm. As the percentage of *T. zygis* in the treatment increased, the average zone of inhibition trended upwards.

3.2. Biofilm Inhibition by Essential Oil Combinations

	Treatment										
	3 (0:10 T:R)	4 (1:9 T:R)	5 (2:8 T:R)	6 (3:7 T:R)	7 (4:6 T:R)	8 (5:5 T:R)	9 (6:4 T:R)	10 (7:3 T:R)	11 (8:2 T:R)	12 (9:1 T:R)	13 (10:0 T:R)
% Inhibition	37.95	49.61	40.63	75.79	50.99	44.68	80.09	67.78	63.08	80.89	70.87
Standard Deviation	65.88	24.93	79.74	49.98	70.99	88.21	9.24	23.80	47.67	8.47	23.29
P-value compared to group 13	0.013	0.001	0.050	0.622	0.147	0.119	0.047	0.607	0.418	0.030	N/A

Table 4. Effectiveness of inhibiting biofilm formation.

Out of the 11 essential oil combinations, groups 12 (90% *T. zygis*, 10% *R. officinalis*) and 9 (60% *T. zygis*, 40% *R. officinalis*) had the highest antibiofilm activities with inhibition percentages of 80.89% and 80.09%, respectively. Demonstrated by the p-values comparing the treatments to group 13, the biofilm inhibition of groups 3, 4, 5, and 12 were all statistically different from group 13.

4. Discussion

The overuse of antimicrobials, specifically antibiotics, has led to great levels of resistance, specifically by UPEC (WHO, 2021), with exacerbation of the problem by biofilm development (Sanchez et al., 2013). *T. zygis* and *R. officinalis* EOs have been researched as antibacterial and antibiofilm treatments, showing great potential (Lagha et al., 2019). This study looked at the two EOs in combination, and were tested for their antibacterial and antibiofilm activities using disk diffusion and crystal violet CFU/mL assay methods, respectively.

The 90% *T. zygis*, 10% *R. officinalis* and 100% *T. zygis*, 0% *R. officinalis* combinations had the highest antibacterial activities, and had numerically higher activities compared to the ciprofloxacin control (Table 3, Figure 5). However, due to the small number of trials conducted of the disk diffusion assay because of resource restraint, the argument could not be made that the EO combinations had statistically higher antibacterial activities compared to the ciprofloxacin. These two combinations were made mainly of *T. zygis* EO, and seeing as *T. zygis* EO has proven to have greater antibacterial capabilities than *R. officinalis* (Lagha et al., 2019), this was an expected outcome. It does, however, say that *R. officinalis* and *T.zygis* do not have additive or synergistic effects because if they did, then the combinations with the highest antibacterial activities would have had more or any *R. officinalis* in it, and would most likely have larger zones of inhibition.

The standard deviations of the data were relatively small, however, this can also be attributed to the small sample size.

The 90% *T. zygis*, 10% *R. officinalis* and 60% *T. zygis*, 40% *R. officinalis* combinations had the highest antibiofilm activities, and did have statistically higher activities compared to the control, which was 100% *T. zygis*, 0% *R. officinalis* (Table 4). These two combinations both had a majority of *T.zygis* compared to *R.officinalis*, but *R. officinalis* has been proven to have greater antibiofilm capabilities compared to *T. zygis* (Lagha et al., 2019), so it can be concluded that there is some sort of enhancing activity occurring between the two essential oils. This enhancing effect would have to be researched further in a more quantifiable manner in order to be categorized, but this study shows promise for the abilities of these two EOs in combination. The standard deviations of the treatments ranged from 8.47 - 79.74, which is a lot of variation. This large range of variation could be attributed to human error, however, if this experiment were to be replicated, the standard deviations should be compared to the results of this study. This variation, however, was accounted for in statistical analysis, so the results of this study are sound.

The intersection between these two assays was the 90% *T. zygis*, 10% *R. officinalis*, as it was the only combination that had the highest antibacterial and antibiofilm activities. By being the only intersection between the two experiments, it was the automatic optimized combination. There were no uncontrolled events that impacted either of the experiments, so all results can be taken as valid.This is a different result than hypothesized, and different than the results from Williams, 2018. This signifies that further research into EO ratios when used in combination is necessary. These results confirm that EO combination therapy is more effective than monotherapy against both bacterial and biofilm growth, meaning that combination therapy should be researched further and could have true potential as alternative treatment.

If this experiment were to be done again, the disk diffusion assay should be replicated several more times, so that better and more valid statistical analysis could be conducted on the values. In addition, ciprofloxacin could be used as another control in the CFU/mL assay, to have a standard of comparison against something that is already on the market as a UTI treatment. Since these results prove that these two essential oils in combination are effective enough to be UTI treatments, further research on this topic could venture into using a 90% *T. zygis*, 10% *R. officinalis* EO combination on pathogenic *E. coli*, small test insects with UPEC, such as *Drosophila melanogaster*, or other types of gram-negative bacteria, such as *Pseudomonas aeruginosa* (*Gram-Negative Bacteria Infections in Healthcare Settings | HAI*). Conducting further experiments with other EO combinations that have the same antibacterial and antibiofilm potentials as *T.zygis* and *R. officinalis* on UPEC would also be valuable, as it would venture into identifying more possible alternative treatments for UPEC-caused UTIs.

5. Conclusion

Treating UTIs caused by UPEC has become increasingly difficult because of high antibiotic resistance levels and biofilms that further this resistance. This finding demonstrated a 90% *T. zygis*, 10% *R. officinalis* EOs as an effective antibacterial and antibiofilm treatment. It also demonstrated that these two EOs enhance each other's effects when used in combination against UPEC. Therefore, it is proposed that this combination be used as an alternative treatment against UTIs in order to spare antibiotic use.

6. Appendix

Appendix 1

Synergism is defined as "interaction of discrete agencies (such as industrial firms), agents (such as drugs), or conditions such that the total effect is greater than the sum of the individual effects" (*Synergism Definition & Meaning*, n.d.). Additive interaction is defined as "a deviation from additivity of the absolute effects of two risk factors" (Ding, 2014). Antagonism is defined as "an interaction between two or more drugs that have opposite effects on the body [...] [that] may block or reduce the effectiveness of one or more of the drugs" (*Drug Antagonism | NIH*, n.d.).

7. Acknowledgements

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References

Algburi, A., Comito, N., & Kashtanov, D. (2016, 11 18). Control of Biofilm Formation: Antibiotics and Beyond. *Applied and Environmental Microbiology*. https://journals.asm.org/doi/pdf/10.1128/AEM.02508-16

Aztreonam Injection. (n.d.). MedlinePlus. Retrieved December 8, 2021, from https://medlineplus.gov/druginfo/meds/a687010.html

Balouiri, M., Sadiki, M., & Ibnsouda, S. K. (2016, 04). Methods for in vitro evaluating antimicrobial activity: A review. *Journal of Pharmaceutical Analysis*, *6*. NCBI. 10.1016/j.jpha.2015.11.005

Bassolé, I., & Juliani, H. (2012, 04 2). Essential Oils in Combination and Their Antimicrobial Properties. *Molecules*. NCBI. 10.3390/molecules17043989

Behrouzi, A., Vaziri, F., & Rad, F. R. (2018, 08 01). Comparative study of pathogenic and non-pathogenic Escherichia coli outer membrane vesicles and prediction of host-interactions with TLR signaling pathways. *BMC Research Notes*. NCBI. 10.1186/s13104-018-3648-3

Bryce, A., Costelloe, C., & Hawcroft, C. (2016, 07 25). Faecal carriage of antibiotic resistant Escherichia coli in asymptomatic children and associations with primary care antibiotic prescribing: a systematic review and meta-analysis. *BMC Infectious Disease*. PubMed. 10.1186/s12879-016-1697-6

Bryce, A., Hay, A. D., & Lane, I. F. (2016, 03 15). *Global prevalence of antibiotic resistance in paediatric urinary tract infections caused by Escherichia coli and association with routine use of antibiotics in primary care: systematic review and meta-analysis*. BMJ. https://www.bmj.com/content/352/bmj.i939

Bukovská, A., Cikoš, Š., & Juhás, Š. (2007). Effects of a Combination of Thyme and Oregano Essential Oils on TNBS-Induced Colitis in Mice. *Mediators of Inflammation*. https://doi.org/10.1155/2007/23296

Cáceres, M., Hidalgo, W., & Stashenko, E. (2020, 03 30). Essential Oils of Aromatic Plants with Antibacterial, Anti-Biofilm and Anti-Quorum Sensing Activities against Pathogenic Bacteria. *Antibiotics*, *9*(4). MDPI. https://doi.org/10.3390/antibiotics9040147

Cho, S.-H., Lim, Y.-S., & Park, M.-S. (2011, 06). Prevalence of Antibiotic Resistance in Escherichia coli Fecal Isolates From Healthy Persons and Patients With Diarrhea. *Osong Public Health and Research Perspectives*, *2*(1), 41-45. Science Direct. https://doi.org/10.1016/j.phrp.2011.05.003

Costerton, J. W., Lewandowski, Z., Caldwell, D. E., Korber, D. R., & Lappin-Scott, H. M. (1995). Microbial biofilms. Annual Review Microbiology. PubMed. 10.1146/annurev.mi.49.100195.003431

Costerton, J. W., Stewart, P. S., & Greenberg, E. P. (1999, 05 21). Bacterial biofilms: a common cause of persistent infections. *Science*. PubMed. 10.1126/science.284.5418.1318

Co-trimoxazole. (n.d.). MedlinePlus. Retrieved December 8, 2021, from https://medlineplus.gov/druginfo/meds/a684026.html

Crystal Violet Assay. (n.d.). University of Kansas Medical Center. https://www2.kumc.edu/soalab/lablinks/protocols/cvassay.htm

Dimitrova, L., Kaleva, M., & Zaharieva, M. M. (2021, 08 4). Prevalence of Antibiotic-Resistant Escherichia coli Isolated from Swine Faeces and Lagoons in Bulgaria. *Antibiotics*, *10*(8). MDPI. https://doi.org/10.3390/antibiotics10080940

Ding, B. (2014). Chapter Four - High-Throughput Genetic Interaction Study. *Between the Lines of Genetic Code*. Science Direct. https://doi.org/10.1016/B978-0-12-397017-6.00004-0

Doterra. (n.d.). *Part 5: Packaging and Storage*. Doterra. https://www.doterra.com/US/en/essential-oil-production-packaging-and-storage-overview

Doxycycline (Oral Route) Description and Brand Names. (n.d.). Mayo Clinic. Retrieved December 7, 2021, from https://www.mayoclinic.org/drugs-supplements/doxycycline-oral-route/description/drg-20068229

Drug Antagonism | NIH. (n.d.). Clinical Info HIV.gov. Retrieved December 5, 2021, from https://clinicalinfo.hiv.gov/en/glossary/drug-antagonism

Flores-Mireles, A., Walker, J., & Caparon, M. (2015, 04 08). Urinary tract infections: epidemiology, mechanisms of infection and treatment options. Nat Rev Microbiol, 13, 269-284. Nature. https://doi.org/10.1038/nrmicro3432

GBD. (2018, 11 10). Global, regional, and national incidence, prevalence, and years lived with disability for 354 diseases and injuries for 195 countries and territories, 1990–2017: a systematic analysis for the Global Burden of Disease Study 2017. *The Lancet*, *392*(10159), 1789-1858. https://doi.org/10.1016/S0140-6736(18)32279-7

High levels of antibiotic resistance found worldwide, new data shows. (2018, January 29). WHO | World Health Organization. Retrieved December 8, 2021, from https://www.who.int/news/item/29-01-2018-high-levels-of-antibiotic-resistance-found-worldwide-new-data-shows

Høiby, N., Bjarnsholt, T., & Moser, C. (2015, 05 1). ESCMID guideline for the diagnosis and treatment of biofilm infections 2014. *Clinical Microbiology and Infection*. Science Direct. https://doi.org/10.1016/j.cmi.2014.10.024

Hölzl-Armstrong, L., Kucab, J. E., Korenjak, M., Lujiten, M., Phillips, D. H., Zavadil, J., & Arlt, V. M. (2019, 11 13). Characterising Mutational Spectra of Carcinogens in the Tumour Suppressor Gene TP53 Using Human TP53 Knock-in (Hupki) Mouse Embryo Fibroblasts. *Methods and Protocols*. 10.3390/mps2040085

Kerekes, E. B., Vidács, A., & Takó, M. (2019, 09 12). Anti-Biofilm Effect of Selected Essential Oils and Main Components on Mono- and Polymicrobic Bacterial Cultures. *Microorganisms*, *7*(9). NCBI. 10.3390/microorganisms7090345

Lagha, R., Abdallah, F. B., AL-Sarhan, B. O., & Al-Sodany, Y. (2019, 03 23). Antibacterial and Biofilm Inhibitory Activity of Medicinal Plant Essential Oils Against Escherichia coli Isolated from UTI Patients. *NCBI*. 10.3390/molecules24061161

Lesovskaya, M. I., & Shaporova, Z. E. (2019). Adaptogenic effects of essential oils: prognosis in vitro and results in vivo. *IOP Conference Series: Earth and Environmental Science*. https://iopscience.iop.org/article/10.1088/1755-1315/315/4/042001/pdf

MacKinnon, M.C., Sargeant, J.M., Pearl, D.L. et al. Evaluation of the health and healthcare system burden due to antimicrobial-resistant Escherichia coli infections in humans: a systematic review and meta-analysis. Antimicrob Resist Infect Control 9, 200 (2020). https://doi.org/10.1186/s13756-020-00863-x

Madappa, T., & Stuart, M. (2019, 02 11). *Which medications in the drug class Antibiotics are used in the treatment of Escherichia coli (E coli) Infections?* Medscape. Retrieved December 7, 2021, from https://www.medscape.com/answers/217485-40715/which-medications-in-the-drug-class-antibiotics-are-used-in-the-treatment-of-escherichia-coli-e-coli-infections

Mahizan, N., Yang, S.-K., Moo, C.-L., Song, A., Chong, C.-M., Chong, C.-W., Abushelaibi, A., Lim, S.-H., & Lai, K.-S. (2019, 06 19). Terpene Derivatives as a Potential Agent against Antimicrobial Resistance (AMR) Pathogens. *Molecules*. NCBI. 10.3390/molecules24142631

Medina, M., & Castillo-Pino, E. (2019). An introduction to the epidemiology and burden of urinary tract infections. Therapeutic advances in urology, 11, 1756287219832172. https://doi.org/10.1177/1756287219832172

Merrem IV (Meropenem): Uses, Dosage, Side Effects, Interactions, Warning. (n.d.). RxList. Retrieved December 8, 2021, from https://www.rxlist.com/merrem-iv-drug.htm

Millezi, A. F., Costa, K., & Oliveira, J. (2019). Antibacterial and anti-biofilm activity of cinnamon essential oil and eugenol. *Clenc. Rural*, *49*(1). https://doi.org/10.1590/0103-8478cr20180314

Neyret, C., Herry, J.-M., Meylheuc, T., & Dubois-Brissonnet, F. (2014, 01 1). Plant-derived compounds as natural antimicrobials to control paper mill biofilms. *Journal of Industrial Microbiology and Biotechnology*, *41*(1), 87-96. https://doi.org/10.1007/s10295-013-1365-4

Nitrofurantoin - Side Effects, Uses, Dosage, Overdose, Pregnancy, Alcohol. (2015, July 23). RxWiki. Retrieved December 8, 2021, from https://www.rxwiki.com/nitrofurantoin

Nji, E., Kazibwe, J., & Hambridge, T. (2021, 02 09). High prevalence of antibiotic resistance in commensal Escherichia coli from healthy human sources in community settings. *Scientific Reports*. Nature. https://www.nature.com/articles/s41598-021-82693-4

Nordmann, P., & Poirel, L. (2019, 11 13). Epidemiology and Diagnostics of Carbapenem Resistance in Gram-negative Bacteria. *Clinical Infectious Diseases*, *69*(7). Oxford Academic. https://doi.org/10.1093/cid/ciz824

Römling, U., & Balsalobre, C. (2012, 10 01). Biofilm infections, their resilience to therapy and innovative treatment strategies. *Journal of Internal Medicine*. Wiley Online Library. https://doi.org/10.1111/joim.12004

Saad, N. Y., Muller, C. D., & Lobstein, A. (2013, 05 09). Major bioactivities and mechanism of action of essential oils and their components. *Flavour and Fragrance Journal*. Wiley Online Library. https://doi.org/10.1002/ffj.3165

Sanchez, C. J., Mende, K., & Beckius, M. L. (2013, 01 29). Biofilm formation by clinical isolates and the implications in chronic infections. *BMS Infect Dis*. PubMed. 10.1186/1471-2334-13-47

Silhayy, T. J. (2015, 12 29). Classic Spotlight: Gram-Negative Bacteria Have Two Membranes. *Journal of Bacteriology*, *198*(2). https://doi.org/10.1128/JB.00599-15

Smieszek, T., & Pouwels, K. B. (2018, 02 27). Potential for reducing inappropriate antibiotic prescribing in English primary care. *Journal of Antimicrobial Chemotherapy*, *73*, ii36-ii43. Oxford Academic. https://doi.org/10.1093/jac/dkx500

Synergism Definition & Meaning. (n.d.). Merriam-Webster. Retrieved December 5, 2021, from https://www.merriam-webster.com/dictionary/synergism

Terahara, F., & Nishiura, H. (2019, 04 23). Fluoroquinolone consumption and Escherichia coli resistance in Japan: an ecological study. https://bmcpublichealth.biomedcentral.com/articles/10.1186/s12889-019-6804-3

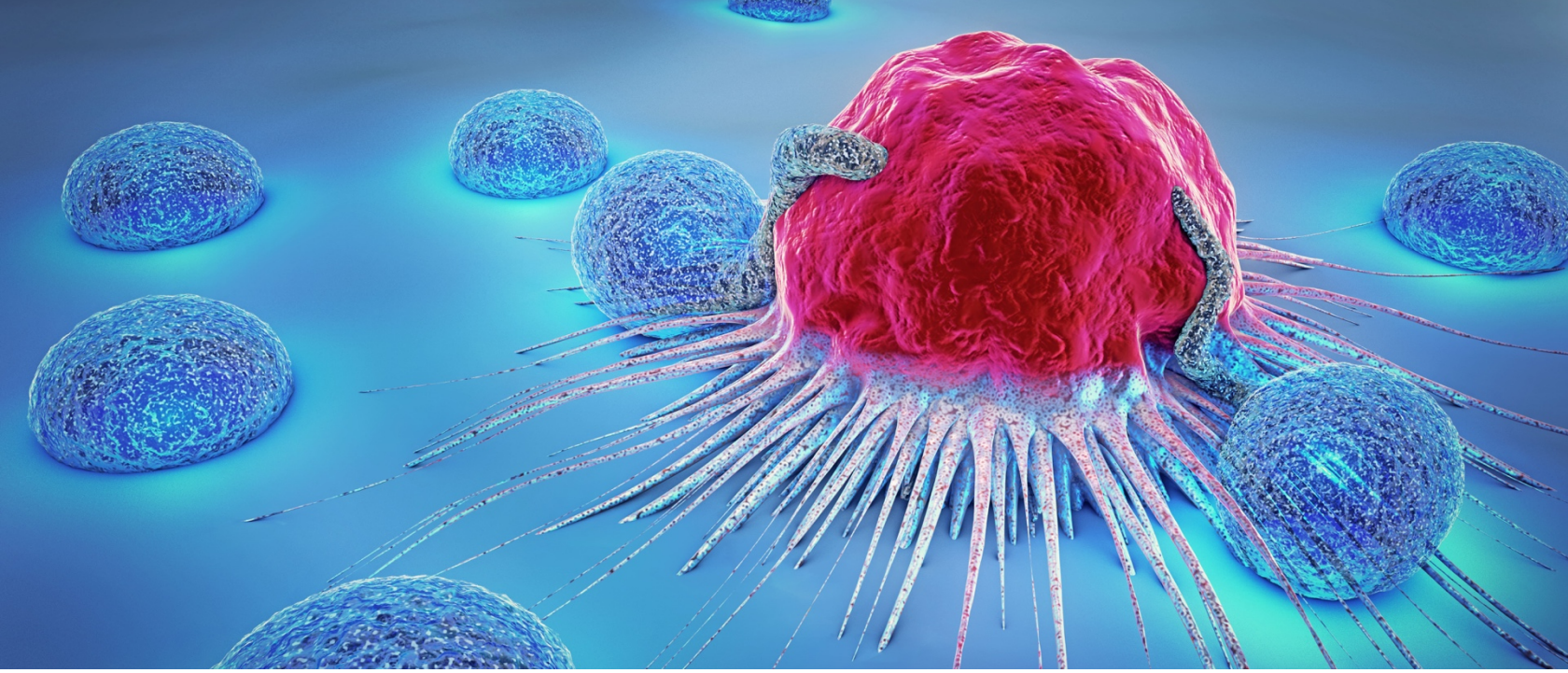
Terlizzi, M. E., Gribaudo, G., & Maffei, M. E. (2017, 08 15). UroPathogenic Escherichia coli (UPEC) Infections: Virulence Factors, Bladder Responses, Antibiotic, and Non-antibiotic Antimicrobial Strategies. *Frontiers in Microbiology*. NCBI. 10.3389/fmicb.2017.01566

WHO. (2021). *Global Antimicrobial Resistance and Use Surveillance System (GLASS) Report*. WHO. https://www.who.int/publications/i/item/9789240027336

Williams, B. E. (2018, 05). *Essential Oils in Combination and Their Effect on E. coli Growth*. Ohio University. https://etd.ohiolink.edu/apexprod/rws_etd/send_file/send?accession=ouashonors1524752078918456&disposition=inline

Wilson, C. (2017, 10 24). *Quantitative and Qualitative Assessment Methods for Biofilm Growth: A Mini-review*. NCBI. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6133255/#R21

Wu, D., Ding, Y., Yao, K., Gao, W., & Wang, Y. (2021, 05 25). https://doi.org/10.3389/fped.2021.670470



3. Developing a Novel Follicular Lymphoma Model to Further Personalize CAR T Cell Treatment

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Introduction

CAR T cell therapies are the newest innovation in leukemia cancer therapies in the last few decades. Follicular cancer, a type of cancer that is treated through CAR T cell therapy, is a form of cancer that presents near blood vessels and, when it transforms, can be aggressive. Although this therapy has led to progress against this condition, it does not always lead to full remission. Furthermore, many experience cytokine storms that cause not only serious discomfort to the patient, but also death and inflammatory conditions. Follicular lymphoma patients can also have different cell ratios in their cancer microenvironment, which can also affect the effectiveness of the treatment. In this study, we created a comprehensive model of the follicular lymphoma microenvironment that can be modified to a patient's unique cell ratio. This model then allows scientists to simulate how CAR T cells will interact with the patient's cancer microenvironment and, as a result, change the secretion rates of varying factors in CAR T cells to increase remission rates in follicular lymphoma patients. In this particular study, we changed the secretion rate of Granzyme A in CAR T cells. Throughout the experiment, we measured the efficiency of the CAR T cell therapy using the cell ratio in the microenvironment, the follicular lymphoma motility, and dendritic cell interactions. Our findings can have applications not only to make adjustments to current treatments of follicular lymphoma, but also to introduce the idea of modifying CAR T cells on a molecular scale to improve CAR T cell treatment efficiency.

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Methods

Our model was formulated using the cell ratio in Burkitt's lymphoma, a lymphoma with a similar cell ratio to follicular lymphoma. The ratio of cells in the simulation were 50 follicular lymphoma cells: 2 dendritic cells: 5 M2 cells: 1 M1 cell: 1 T regulatory cell, as shown in Figure 1 and in Figures 2 and 3 with the latter showing the microenvironment as demonstrated in CompuCell3D [1]. The Table 1 and the Figure 1 also shows the cell ratio used in the CompuCell3D simulation. The cell ratio, however, varies from patient to patient and was obtained from an average of 22 samples [2]. The secretion rate of the varying factors for each cell type was estimated using the concentration of different factors in the anaplastic lymphoma, which is similar to the follicular lymphoma microenvironment without any CAR T cell involvement [3]. Then, the concentration of these factors was divided by the number of cells that secrete that particular factor in order to get the secretion rate of each factor for each cell type. It should be noted that there were other factors and cell types present in the study that were referenced; however, due to their minimal impact or small concentration in the microenvironment, they were excluded from the simulation. These values are listed in Table 1 along with information regarding how the different factors affect the lymphoma microenvironment.

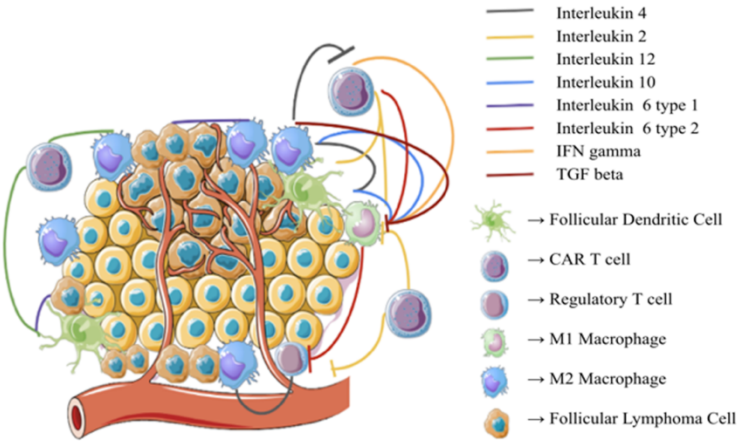


Figure 1. This figure demonstrates the follicular lymphoma microenvironment simulated in CompuCell3D. The diagram depicts the exact ratio of cells used in the simulation and how cells interact through factors, with the straight lines indicating excitatory pathways and ones with a vertical line at the end of them an inhibitory pathway.

Table 1 tabulates the various factors involved in this computation as well as the secretion rates of these factors by the cells in the simulation. These values were calculated by dividing the environmental concentration of these factors by the number of cells in the computation that secrete said factor. It was assumed that each cell type that secretes the factor has the same secretion rate, as there was no information to be found from the scientific community on the specific secretion rates of factors by the cells in the follicular lymphoma microenvironment [13].

Factor Table			
Factor	Field Name	Properties	Secretion Rate
Interleukin 6	IL 6 type 1	Reduces number of migratory immune cells to lymphoma site [4], [5]	0.0385 per M2, Treg, and Intratumoral dendritic cell [3]
Hedgehog ligand	Hh	Prevents lymphoma apoptosis [6], [7]	0.05 per MSC, Intratumoral dendritic, and reticular cell [3]
Interleukin 6	IL 6 type 2	Increases number of migratory immune cells to lymphoma site [8], [9]	0.27 per M1 cell [3]

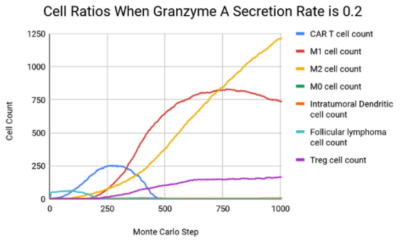
Interleukin 2	IL2	Promotes NK and T cells activation and cell growth [10], [11]	1.89 per CAR T cell [3]
Interleukin 4	IL4	Decreases CD8+ cell growth; increases follicular lymphoma cell growth; increased stromal cell proliferation; regulated lineage of DCs in microenvironment [12]	0.3695 per M2 and Treg cell [3]
Interleukin 10	IL10	Inhibits IL2 and IFN gamma production; Downregulates M1 activity and inhibits expression of CD80 and CD86 which are present on M1 cells which interact with CD28 on CAR T cells and increase CAR T cell activity [13]–[17]	0.097 per M2 and Treg cell [3]
Interleukin 12	IL12	Upregulates IFN gamma production in T and NK cells and increases T cell exhaustion; increases treg activity [18]	0.0765 per Intratumoral dendritic, M2, and Follicular cell [3]
IFN gamma	gamma	Decreases volume and spontaneous creation of lymphomas [16], [19]	0.3045 per CAR T cell [3]
TGF beta	beta	Increased stromal and endothelial cell Infiltration [20]–[23]	0.1165 per M2 cell [3]
CCL19, CCL21, CXCL12	FRsecretion	Increase interactions with T zone which allows for more naïve T and B cells to become assimilated into environment as immunosuppressants and increases B cell migration to area [24], [25]	0.05 per Reticular cell [3]
Interleukin 15	IL15	Increase STAT-5 FL activation [26], [27]	0.0555 per MR and Intratumoral dendritic cell [3]
TNF alpha	alpha	Increases FRC differentiation and increases M2 polarization [29]	0.0585 per Intratumoral dendritic, follicular, and MSC cell [3]
CXCL1, CXCL2, and CXCL12	MSCsecretion	Increase cancer proliferation as well as possibility of metastasis [17]	0.05 per MSC cell [3]

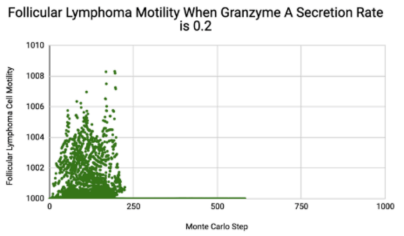
There were several aspects of follicular lymphoma that were excluded from the model because it would require an environment larger than the scope of the model, or it was not possible because there is not enough information available. The T-zone within lymph nodes is one aspect of the microenvironment that was excluded. Reticular cells produce factors CCL19, CCL21, and CXCL12, which cause increased interactions between the T-zone B cells and therefore the follicular lymphoma cells [24]. In lymphomas that are resistant to CAR T cell therapy, genes FADD, BID, CASP8, and TNFRSF10B were deplete and gained genes that resist cell death such as CFLAR, BIRC2, and TRAF2. Lymphomas typically gain these genes from severe

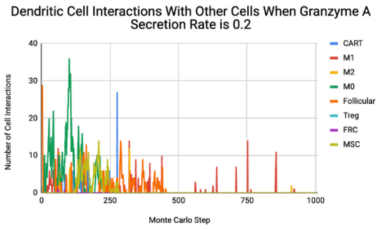
inflammation or stress in the microenvironment caused by CAR T cell therapy [30]. CAR T cell resistant cancers can also take the form of cancer stem cells that can be undifferentiated, allowing them to change the antigens they present on their cell membrane [31]. This phenomenon is called lineage switching-mediated target antigen loss and can render CAR T cell therapy useless [31]. This environment impacts the CAR T cell as it results in diminished cytotoxicity, minimal T cell differentiation, and aerobic glycolysis, Thus, when this occurred, it decreased the cell growth of CAR T cells. This phenomenon was taken into account in the model by setting a threshold of interleukin 6 from M1 cells that can cause follicular lymphoma CAR T cell resistance.

Results

Cell ratios can be used to monitor the status of the microenvironment and the success of the CAR T cell therapy. In Figure 2a, the cell population of a certain cell type are monitored over the duration of the experiment, which is 1000 Monte Carlo steps with each Monte Carlo step equal to 30 minutes in real time. In figure 2b we monitored cell motility of lymphoma cells are directly correlated with metastasis and we found that at a monte carlo step of 238 all cell motility in lymphoma cells went back to their baseline motility. In Figure 2c, we modeled dendritic cell interactions which are vital for the status of the microenvironment. Dendritic cells vary in their activity based on their interactions with other cells. For instance, when dendritic cells interact with pro tumor cells, it causes the dendritic cell to secrete pro-tumor factors such as IL12 and TNF alpha [32]. When dendritic cells interact with anti-tumor cells, the dendritic cells secrete anti- tumor factors such as IL2.







Discussion

In this study, we explored how modifying the Granzyme A secretion rate in CAR T cells can improve patient conditions through CompuCell modeling; however, this approach can be used for the patient's unique cell ratio and can vary different secretion rates in the CAR T cell to improve remission. We could not demonstrate the full complexity of this microenvironment, but we included the parts of the microenvironment most impactful to follicular lymphoma growth. The importance of a cell or factor was determined by its concentration and the significance of its impact on the microenvironment.

From the results, we concluded that 0.2 pg per monte carlo step relative to the other factors is the Granzyme A secretion rate of CAR T cells that was the most beneficial, as it decreased the follicular lymphoma population significantly without creating a highly pro-inflammatory microenvironment. It also caused decreased cell motility in follicular lymphoma cells, which decreased the likelihood of metastasis, or the state in which cancer travels to other parts of the body.

However, this Granzyme A secretion rate is specific to the cell ratio used in the computational representation of the follicular lymphoma microenvironment. This model that we have created of this microenvironment can be altered based on the cell ratios of individual patients, allowing for a further specialized and more effective treatment as shown in other studies. Others have done work predicting how

cancer microenvironments will react to treatments, and our study provides an in-depth demonstration of how follicular lymphoma will respond based on the cell ratio. Changes to the secretion rates of these factors can be achieved by changing the polyadenylation to the mRNA sequences that code for granzyme and granzyme A in the CAR T cell. Our findings demonstrate how CAR T cell therapy can be further specialized for treatment success.

References

Header Image:
Krebs: Chronik des fortschritts. (2019, June 23). Pharma Fakten. https://pharma-fakten.de/news/787-krebs-chronik-des-fortschritts/

[1] M. H. Swat, G. L. Thomas, J. M. Belmonte, A. Shirinifard, D. Hmeljak, and J. A. Glazier, "Multi-Scale Modeling of Tissues Using CompuCell3D," *Methods Cell Biol*, vol. 110, pp. 325–366, Jan. 2012, doi: 10.1016/B978-0-12-388403-9.00013-8.
[2] M. Granai *et al.*, "Immune landscape in Burkitt lymphoma reveals M2-macrophage polarization and correlation between PD-L1 expression and non-canonical EBV latency program," *Infect Agent Cancer*, vol. 15, no. 1, May 2020, doi: 10.1186/s13027-020-00292-w.
[3] F. Knörr *et al.*, "Blood cytokine concentrations in pediatric patients with anaplastic lymphoma kinase-positive anaplastic large cell lymphoma," *Haematologica*, vol. 103, no. 3, pp. 477–485, Feb. 2018, doi: 10.3324/haematol.2017.177972.
[4] D. de Jong and T. Fest, "The microenvironment in follicular lymphoma," *Best Pract Res Clin Haematol*, vol. 24, no. 2, pp. 135–146, Jun. 2011, doi: 10.1016/J.BEHA.2011.02.007.
[5] E. Bettelli *et al.*, "Reciprocal developmental pathways for the generation of pathogenic effector TH17 and regulatory T cells," *Nature*, vol. 441, no. 7090, pp. 235–238, May 2006, doi: 10.1038/nature04753.
[6] G. Casella *et al.*, "IL4 induces IL6-producing M2 macrophages associated to inhibition of neuroinflammation in vitro and in vivo," *Journal of Neuroinflammation*, vol. 13, no. 1, Jun. 2016, doi: 10.1186/s12974-016-0596-5.
[7] G. Basile Carballo, J. Ribeiro Honorato, G. Pinto Farias de Lopes, and T. Cristina Leite de Sampaio Spohr, "A highlight on Sonic hedgehog pathway", doi: 10.1186/s12964-018-0220-7.
[8] C. Dierks *et al.*, "Essential role of stromally induced hedgehog signaling in B-cell malignancies," *Nature Medicine*, vol. 13, no. 8, pp. 944–951, 2007, doi: 10.1038/nm1614.
[9] K. Nakajima *et al.*, "A central role for Stat3 in IL-6-induced regulation of growth and differentiation in Ml leukemia cells," *The EMBO Journal*, vol. 15, no. 14, pp. 3651–3658, 1996.
[10] Y. Yamanaka, K. Nakajima, T. Fukada, M. Hibi, and T. Hirano, "Differentiation and growth arrest signals are generated through the cytoplasmic region of gp130 that is essential for Stat3 activation," *EMBO Journal*, vol. 15, no. 7, pp. 1557–1565, Apr. 1996, doi: 10.1002/j.1460-2075.1996.tb00500.x.
[11] T. Miyazaki *et al.*, "Three Distinct IL-2 Signaling Pathways Mediated by bcl-2, c-myc, and lck Cooperate in Hematopoietic Cell Proliferation," 1995.
[12] S. H. Ross and D. A. Cantrell, "Signaling and Function of Interleukin-2 in T Lymphocytes," *Annu Rev Immunol*, vol. 36, pp. 411–433, Apr. 2018, doi: 10.1146/annurev-immunol-042617-053352.
[13] V. Spieler *et al.*, "Targeting interleukin-4 to the arthritic joint," *Journal of Controlled Release*, vol. 326, pp. 172–180, Oct. 2020, doi: 10.1016/J.JCONREL.2020.07.005.
[14] J. Cortes and R. Kurzrock, "Interleukin-10 in non-Hodgkin's lymphoma," *Leukemia and Lymphoma*, vol. 26, no. 3–4. Informa Healthcare, pp. 251–259, 1997. doi: 10.3109/10428199709051774.
[15] J. K. Riley, K. Takeda, S. Akira, and R. D. Schreiber, "Interleukin-10 receptor signaling through the JAK-STAT pathway. Requirement for two distinct receptor-derived signals for anti-inflammatory action," *Journal of Biological Chemistry*, vol. 274, no. 23, pp. 16513–16521, Jun. 1999, doi: 10.1074/jbc.274.23.16513.
[16] A. H. S-y, S. H-y Wei, A. L-f Mui, A. Miyajima, and K. W. Moore, "Functional Regions of the Mouse Interleukin-10 Receptor Cytoplasmic Domain," 1995. [Online]. Available: https://journals.asm.org/journal/mcb
[17] M. A. Meraz and † J Michael White, "Targeted Disruption of the Stat1 Gene in Mice Reveals Unexpected Physiologic Specificity in the JAK-STAT Signaling Pathway," 1996.
[18] S. J. Rodig *et al.*, "Disruption of the Jak1 Gene Demonstrates Obligatory and Nonredundant Roles of the Jaks in Cytokine-Induced Biologic Responses," 1998.
[19] Z.-Z. Yang *et al.*, "IL-12 upregulates TIM-3 expression and induces T cell exhaustion in patients with follicular B cell non-Hodgkin lymphoma," *The Journal of Clinical Investigation*, vol. 122, 2012, doi: 10.1172/JCI59806.
[20] F. Castro, A. P. Cardoso, R. M. Gonçalves, K. Serre, and M. J. Oliveira, "Interferon-gamma at the crossroads of tumor immune surveillance or evasion," *Frontiers in Immunology*, vol. 9, no. MAY. Frontiers Media S.A., May 04, 2018. doi: 10.3389/fimmu.2018.00847.
[21] P. A. Guerrero and J. H. McCarty, "TGF-β Activation and Signaling in Angiogenesis," in *Physiologic and Pathologic Angiogenesis - Signaling Mechanisms and Targeted Therapy*, InTech, 2017. doi: 10.5772/66405.
[22] B. Tirado-Rodriguez, E. Ortega, P. Segura-Medina, and S. Huerta-Yepez, "TGF-β: An important mediator of allergic disease and a molecule with dual activity in cancer development," *Journal of Immunology Research*, vol. 2014, 2014, doi: 10.1155/2014/318481.
[23] M. Bakkebø, K. Huse, V. I. Hilden, E. B. Smeland, and M. P. Oksvold, "TGF-β-induced growth inhibition in B-cell lymphoma correlates with Smad1/5 signalling and constitutively active p38 MAPK," *BMC Immunology*, vol. 11, Dec. 2010, doi: 10.1186/1471-2172-11-57.
[24] Z. Z. Yang *et al.*, "TGF-β upregulates CD70 expression and induces exhaustion of effector memory T cells in B-cell non-Hodgkin's lymphoma," *Leukemia*, vol. 28, no. 9, pp. 1872–1884, 2014, doi: 10.1038/leu.2014.84.
[25] F. Mourcin *et al.*, "Stromal cell contribution to human follicular lymphoma pathogenesis," 2012, doi: 10.3389/fimmu.2012.00280.

[26] P. Amé -Thomas *et al.*, "Human mesenchymal stem cells isolated from bone marrow and lymphoid organs support tumor B-cell growth: role of stromal cells in follicular lymphoma pathogenesis," 2007, doi: 10.1182/blood-2006-05.
[27] A. Mishra, L. Sullivan, and M. A. Caligiuri, "Molecular pathways: Interleukin-15 signaling in health and in cancer," *Clinical Cancer Research*, vol. 20, no. 8, pp. 2044–2050, Apr. 2014, doi: 10.1158/1078-0432.CCR-12-3603.
[29] J. A. Johnston *et al.*, "Tyrosine phosphorylation and activation of STAT5, STAT3, and Janus kinases by interleukins 2 and 15," 1995.
[30] J. Cheng *et al.*, "Understanding the Mechanisms of Resistance to CAR T-Cell Therapy in Malignancies," *Frontiers in Oncology*, vol. 9. Frontiers Media S.A., Nov. 21, 2019. doi: 10.3389/fonc.2019.01237.
[31] Z. Yu, T. G. Pestell, M. P. Lisanti, and R. G. Pestell, "Cancer stem cells," *International Journal of Biochemistry and Cell Biology*, vol. 44, no. 12. Elsevier Ltd, pp. 2144–2151, 2012. doi: 10.1016/j.biocel.2012.08.022.
[32] F. Veglia and D. I. Gabrilovich, "Dendritic cells in cancer: the role revisited," *Current Opinion in Immunology*, vol. 45. Elsevier Ltd, pp. 43–51, Apr. 01, 2017. doi: 10.1016/j.coi.2017.01.002.



4. The Ever-Changing Brain: Daily Use of Digital Technology and Its Effects on Neuroplasticity

Wan 'Aliyah, Global Jaya School

Fictional stories as we know them are fabricated by novelists who have the power to create and control the plot. Almost like fiction, the human brain possesses an incredible ability to rewire itself: it is the author that writes and the pencil that draws its version of literature.

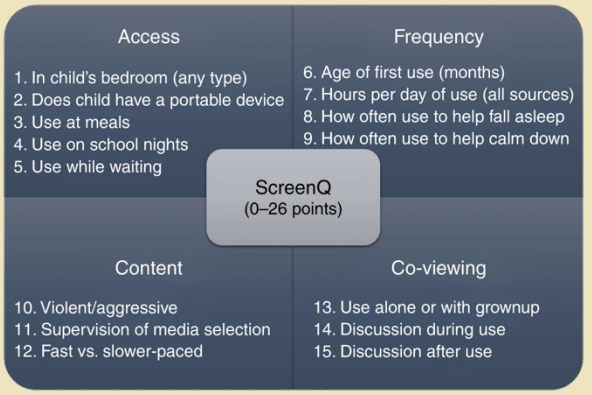
This phenomenon is referred to as neuroplasticity, which is the capability of neural networks inside the brain to adapt, reorganise and change their behaviour in response to new experiences, information, stimuli or damage [1]. It is the reason bad habits remain ingrained in the brain and that even valuable skills can be lost over time—indeed, a "use it or lose it" system.

Structural neuroplasticity involves reshaping individual neurons and changing their physical structure due to learning and experience. The neurons in the brain send and receive electrical impulses connected by a synapse acting as a bridge between one neuron and another. While repetition and practice strengthen these existing synapses connections within the brain, new experiences and skills can form new ones—this is known as neurogenesis [2]. In contrast, weakened synaptic connections that remain unused in the brain are eliminated during synaptic pruning to keep its processes more efficient [2]. Due to its malleability, the brain can easily be influenced, trained and controlled by external factors, including an individual's lifestyle and habits [3].

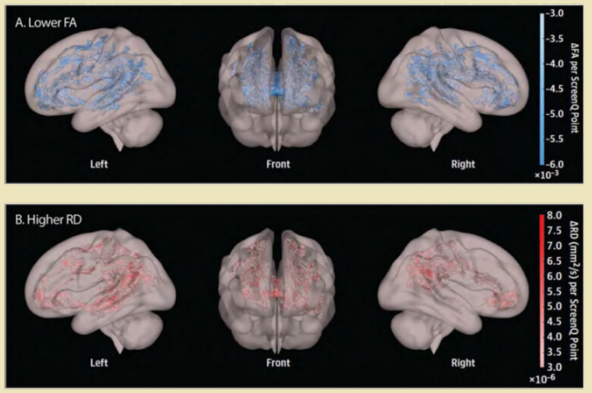
Researchers examine neuroplasticity through neuronal changes in the brain using different types of magnetic resonance imaging (MRI). Neuronal changes can be categorised by grey and white matter in the brain. They are responsible for the conduction, processing, and transmission of information as well as the speed of the impulse. They may influence MRI signals, making it the perfect modality to measure neuroplasticity [4].

In a world where technology is a significant aspect of personal lives, it has become a habit and a lifestyle that many cannot recall a life without instant connectivity to the internet. Modern cognition such as navigation, calculation, communication and memorisation rely on technology. From accessibility to productivity, it provides all the facets that facilitate daily activities by either enhancing or extending mental capabilities. Everyday exposure to technological devices, media and the internet influences the way individuals access and engage with information. It stimulates alteration in their brain activity to a profound degree. The biological changes induced by exposure to technology have led researchers to examine the response and transformation that occur in human brains and their cognitive ability.

Technology and Language in Developing Brains



Conceptual diagram of the ScreenQ measure and its four domains. <5>



Diffusion tensor magnetic resonance imaging of the brain in preschoolers. <6>

An area of neuroplastic interest is whether intensive digital media usage affects the development of processes linked with language. This study estimates white-matter integrity in the brain of preschool children using a standardised 15-item screening tool known as ScreenQ. The ScreenQ scores were then statistically correlated with the children's diffusion tensor MRI measurement and cognitive test scores.

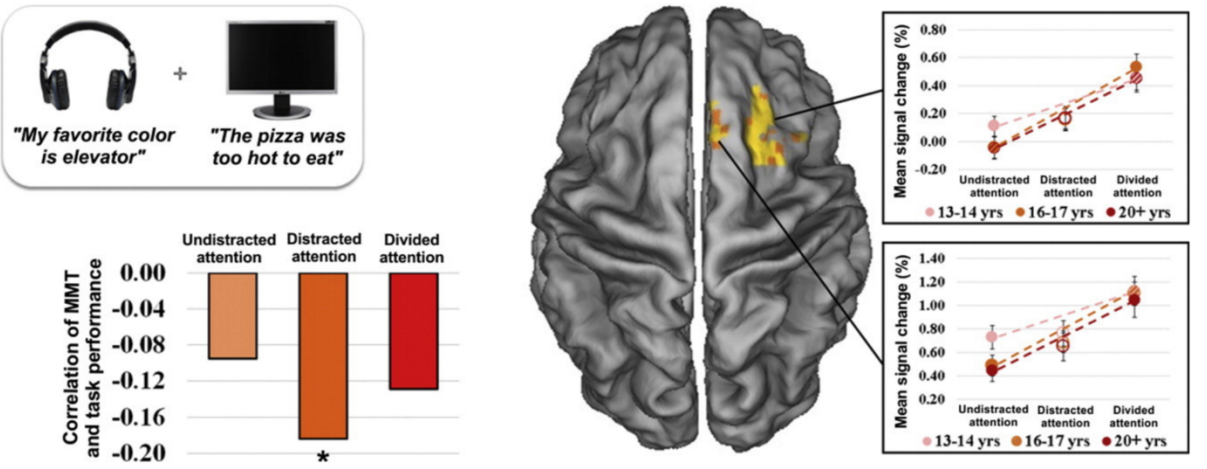
The overall white matter volume results showed a significant correlation between ScreenQ scores, lower fractional anisotropy (FA) and higher radial diffusivity (RD). The FA and RD indicate the fibre tract in the whole-brain images, suggesting a link between intensive early childhood digital media use and poorer microstructural integrity of white matter tracts. It is most prominent in areas involved in language and the comprehension of speech: the Broca and Wernicke areas in the brain. The researchers observed lower executive functions and literacy abilities in the study despite matching age and average household income. As a result of digital media usage, the fibre tracts in language areas of the children's brain were not developed to their full extent [6].

Media Multi-Tasking: Attention and Memory

Technological accessibility has blurred the lines between online and offline activities and allows us to perform multiple concurrent tasks. Skimming past information with superficial interest and shifting from one activity to another could result in short-term memories. This habit trains the brain to diminish the internalisation and retention of information. That is because the brain does not actively engage in the processes necessary for forming long-term memories [7]. At the same time, this also affects an individual's attentional functioning.

A study found that those who engage in frequent media multi-tasking (MMT) in their daily lives experience tend to perform poorer in several cognitive tasks, especially for sustained attention [7]. In the study, 149

adolescents and young adults were assigned speech-listening and reading assignments that required them to maintain attention in the presence of distractor stimuli. The participants' performance was measured using functional magnetic resonance imaging (fMRI).



Media multitasking is associated with distractibility and increased prefrontal activity in adolescents and young adults <8>

Based on the study, a higher media multi-tasking score was associated with poor performance and increased brain activity in the right prefrontal regions. Despite the increased activity, the performance was lower than light media multi-taskers. The findings suggest that heavy media multi-taskers require a greater cognitive effort to maintain concentration when faced with distractor stimuli. In terms of structure, higher internet usage and heavy media multi-tasking levels are related to decreased grey matter in prefrontal regions. The reduction in grey matter possibly indicates the unpracticed skill in maintaining goals and focusing in the face of a distraction [9].

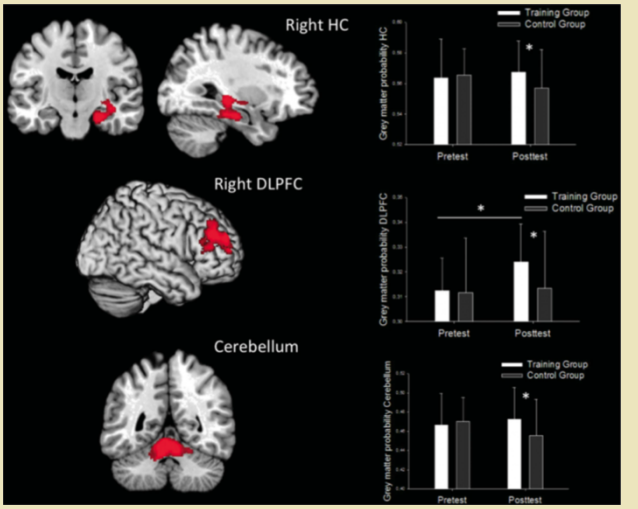
Internet Usage: Changes in Reward Pathways

On top of it all, ever wondered why switching off social media may be challenging? According to research, these social media habits might be due to neuroplasticity [10]. When an urge is satisfied, dopamine, a neurotransmitter that gives pleasure, is released. This chemical is essential to neuroplastic change as it assists in building neural connections that reinforce that particular habit. This reward pathway motivates people to seek and repeat the activity. In this case, the reward pathway shift could lead to addiction to the internet and social media [11].

Positive Neuroplasticity

Despite these potentially harmful brain-health effects, studies also show that video games and digital technological experiences provide opportunities for cognitive sharpening through brain-strengthening neural exercises [12].

The Benefits of Video Games



Brain regions showing a significant group (training vs control) × time (pre vs post-test) <13>

Video games can be both detrimental and beneficial depending on the time spent playing and the training of skills involved. A study in 2013 conducted by Germany's Max Planck Institute for Human Development and St. Hedwig-Krankenhaus investigated the effects of video games on the brain. Results that were measured using MRI revealed that playing Super Mario 64 increases grey matter volume in the right hippocampus (HC), right dorsolateral prefrontal cortex (DLPFC) and cerebellum. These brain regions are responsible for memory formation, strategic planning, spatial navigation, and fine motor skills. The changes in grey matter volume were more pronounced in participants who were eager to play, reflecting the role of motivation in the learning process as it builds and strengthens synapses connections [14]. Thus, the cognitive demands in mastering video games have led to the formation of new synaptic connections in brain sites responsible for spatial navigation, planning and decision making [2]. It is advantageous to encourage positive changes in neural networks such as this.

Improving Neuroplasticity

The neuroplastic change consists of five components: challenge and novelty, intention, specific attention, repetition and intensity, and time of the practice [15]. Begin by selecting an activity that is new, challenging and relevant. Committing to the engagement of new healthy exercises and habits will lead to favourable long-lasting neuroplasticity changes. The list of activities to choose from is massive! It may include learning something new such as an instrument or a new language, personal expression in different forms of creativity, travelling and exploring new places, meeting new people and much more. Even planning a healthy diet or workout routine or fixing an unhealthy sleeping schedule is one step closer to a positive change. Don't forget to get plenty of sleep as well, as it facilitates brain plasticity growth and development.

As neuroplasticity continues to promote both positive and negative changes, it emphasises the importance of personal balance and repetition to maintain essential and practical skills in the future. Exposure to digital media and technology is inevitable. Although the focus of neuroplasticity was centred on the negative aspects of technology, numerous positive elements have yet to be discussed. Taking advantage of the efficiencies of new technologies could prove to be a future asset. Rewiring the human brain and its plasticity depends on an individual and the lifestyle they choose to follow.

References

Header Image:

Alimran. (2020, June 08). Neuroplasticity and CNS reorganization: Alimran Medical Center. Retrieved June 14, 2022, from https://alimranmed.com/2020/06/08/neuroplasticity-and-cns-reorganization/

[1] Rugnetta, M. (n.d.). Neuroplasticity. Retrieved June 13, 2022, from https://www.britannica.com/science/neuroplasticity

[2] Gamma, E. (2021, March 24). What is brain plasticity? Retrieved June 13, 2022, from https://www.simplypsychology.org/brain-plasticity.html

[3] Cherry, K. (2022, February 18). How brain neurons change over time from life experience. Retrieved June 13, 2022, from https://www.verywellmind.com/what-is-brain-plasticity-2794886

[4] Reid, L., Boyd, R., Cunnington, R., & Rose, S. (2015, December 29). Interpreting intervention induced neuroplasticity with fmri: The case for Multimodal Imaging Strategies. Retrieved June 13, 2022, from https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4709757/#:~:text=One%20very%20popular%20modality%20used,a%20comparison%20or%20resting%20state.

<5> Hutton, J et al., (2020, February 12). A novel, composite measure of screen-based media use in young children (ScreenQ) and associations with parenting practices and Cognitive Abilities. Retrieved June 13, 2022, from https://www.nature.com/articles/s41390-020-0765-1#:~:text=ScreenQ%20is%20a%2015%2Ditem,scores%20reflecting%20greater%20non%2Dadherence.

[6] The impact of the Digital Revolution on human brain and behavior: Where do we stand? (2022, April 01). Retrieved June 13, 2022, from https://www.tandfonline.com/doi/full/10.31887/DCNS.2020.22.2/mkorte

[7] Anderson, I. (2013). Thinking in 140 Characters: The Internet, Neuroplasticity, and Intelligence Analysis. Retrieved from https://digitalcommons.usf.edu/cgi/viewcontent.cgi?article=1268&context=jss

<8> Salmela, V., Hietajärvi et al., (2016, July 01). Media multitasking is associated with distractibility and increased prefrontal activity in adolescents and young adults. Retrieved June 14, 2022, from https://www.sciencedirect.com/science/article/abs/pii/S1053811916300441

[9] Firth, J et al., (2019, May 06). The "Online Brain": How the Internet may be changing our cognition. Retrieved June 13, 2022, from https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6502424/

[10] What social media is really doing to your brain. (2017, October 10). Retrieved June 13, 2022, from https://www.nib.com.au/the-checkup/what-social-media-is-really-doing-to-your-brain

[11] 3 ways negative neuroplasticity hurts you. (2017, November 12). Retrieved June 13, 2022, From https://thebestbrainpossible.com/mental-health-neuroplasticity-brain-habits-negative/

[12] Small, G et al., (2020, June). Brain health consequences of digital technology use. Retrieved June 13, 2022, from https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7366948/?report=classic

<13> Watson, A. (n.d.). Super mario changes your brain. Retrieved June 13, 2022, from https://www.nationalelfservice.net/diagnosis/brain-imaging/super-mario-changes-your-brain/

[14] Video game playing found beneficial for the brain. (2013, November 01). Retrieved June 13, 2022, from https://www.kurzweilai.net/video-game-playing-found-beneficial-for-the-brain

[15] Call, M. (2019, August 08). Neuroplasticity: How to use your brain's malleability to improve your well-being. Retrieved June 14, 2022, from https://accelerate.uofuhealth.utah.edu/resilience/neuroplasticity-how-to-use-your-brain-s-malleability-to-improve-your



5. Magnesium Fact Check: How is Magnesium Helpful for the Bone and the Muscle?

Jihyeon Choi & Eun Gyul Hwang, Global Jaya School

Have you ever experienced shaking in the muscles below your eye? It is called eye twitching and can be caused by magnesium deficiency and persistent stress, which leads to an imbalance in magnesium levels. This stops your eyelid muscles from relaxing properly, so your eyelid muscle twitches. There are many other similar medical phenomena associated with magnesium. In fact, magnesium plays a critical role in our body [3].

Magnesium is a chemical element with the symbol Mg and atomic number 12. It is an alkaline-earth metal of group 2 on the periodic table and the lightest structural metal. In addition, it is the eighth-most abundant element in Earth's crust, making up approximately 2.5 percent [7].

This element is an essential mineral for the body. Its compounds are widely used in construction and medicine. Furthermore, magnesium is one of the elements essential to all cellular life and processes [7].

A healthy body needs proper nutrition; the required magnesium consumption depends on age and sex. For example, 14 to 18 year-old teen girls are required to consume 360 mg of magnesium per day while boys require 410 mg per day. Consuming the recommended amount is important because magnesium supports many bodily functions including nerve signals, muscle contraction, energy production, blood pressure, and more [10].

Bone consists of 60% magnesium. The mineral also contributes to bone-building cells and the parathyroid hormone, which balances calcium levels. In fact, it has been found that a higher intake of magnesium resulted in greater bone mineral density. A Women's Health Initiative study of 73,684 postmenopausal women found that a lower magnesium intake was associated with lower bone mineral density throughout the entire body [8].

Magnesium is also involved in bone growth and repair, directly modifying the structure and the size of the bone crystal. Furthermore, magnesium reduces and controls the concentrations of parathyroid hormone (PTH) and vitamin D, which are highly relevant to bone disorders [13]. In summary, magnesium is a mineral that plays an important role in maintaining healthy bones by contributing to increased bone density and preventing the onset of osteoporosis.

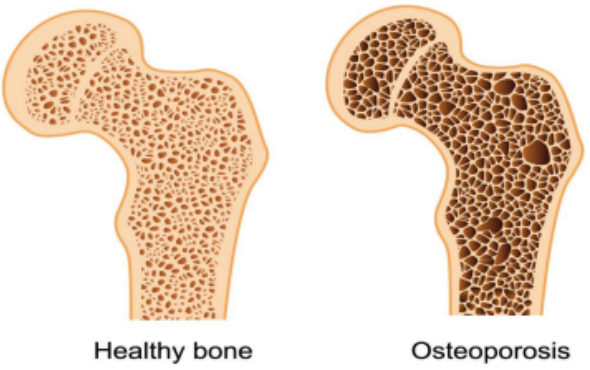


Photo of healthy bone and bone with osteoporosis <14>

Since magnesium works closely with calcium, it is important to have an appropriate ratio of both in order for them to be effective. A good rule of thumb is a 1:2 magnesium and calcium ratio. For example, if you take 400 mg of magnesium, you should also take 800 mg of calcium [11].



Lower leg with muscle cramp <6>

In addition, magnesium is helpful for the muscular system. It helps regulate muscle contraction and works as a natural calcium blocker to help muscles relax. Inside the muscle, calcium binds proteins, such as troponin C and myosin, to alter the shape of the two proteins, therefore causing contraction. In contrast, magnesium targets the same binding sites as calcium in order to relax the muscle. Therefore, if the body does not have the proper amount of magnesium to compete with calcium, the muscles contract too much and start to cramp or spasm [12].

Magnesium helps the muscle by lessening the build-up of lactic acid, the chemical that causes muscular tension. Magnesium does this by allowing the muscle to get the oxygen that it needs [2]. Twitches, tremors, and muscle cramps are signs of magnesium deficiency. While occasional twitches are common, you should see your doctor if your symptoms persist. Muscle cramps are sudden, involuntary contractions that occur in various muscles. These contractions are often painful and can affect different muscle groups. Commonly affected muscles include those in the back of the leg and thigh [4].

Fatigue is a condition characterized by physical or mental exhaustion and is another symptom of magnesium deficiency. Typically, rest will resolve the symptoms; however, severe or persistent fatigue may be a sign of a health problem. Myasthenia gravis is another sign of magnesium deficiency. Myasthenia gravis most commonly affects the muscles that control most ports of the body including the eyes and eyelids, facial expressions, chewing, swallowing, and speaking [1].

There are different types of magnesium. Magnesium chloride is a magnesium salt that includes chlorine. This compound is easily absorbed in the digestive tract. It is usually taken in the form of a capsule or tablet but it is also contained in products like lotions and ointments. This type of magnesium is generally used for relaxing muscles. Another type that is useful to muscular health is magnesium sulfate. It is a form of magnesium that is combined with sulfur and oxygen. Magnesium sulfate can be consumed through the digestive system or it can be dissolved in bath water and absorbed by the skin. It is sometimes contained in skin care lotions. This magnesium is generally used for achy muscles and relieves stress. There are also magnesium citrate, magnesium oxide, magnesium lactate, magnesium malate, magnesium taurate,

magnesium L-threonate, magnesium glycinate, and magnesium orotate, which are helpful for digestive disorders, mental diseases, and cardiovascular diseases [4].

The recommended daily amount of magnesium is 300 mg to 500 mg for adults [11]. Sometimes people can take in too much magnesium in antacids or laxatives. Taking more than the recommended dose can cause vomiting, diarrhea, abdominal cramps, and irregular heartbeat in extreme cases [9].

References

Header Image:
Fungsi Magnesium, Mengatur Fungsi Organ Hingga Cegah Depresi. (2019, November 19). [Photograph]. SehatQ. https://www.sehatq.com/artikel/pentingnya-manfaat-magnesium-bahkan-bisa-cegah-depr esi

[1] BSc, A. A., PhD. (2022, April 12). *7 Signs and Symptoms of Magnesium Deficiency*. Healthline. https://www.healthline.com/nutrition/magnesium-deficiency-symptoms#TOC_TITLE_H DR_5
[2] Escaping Gravity Ltd. Collaborator. (2022, May 9). *Magnesium for muscle tension—how it works and how to choose the right type of supplement*. BetterYou. https://betteryou.com/blogs/health-hub/how-magnesium-supplements-can-relieve-muscle -tension
[3] *Eye twitching due to magnesium deficiency | Biolectra® Magnesium*. (2022, March 14). Biolectra Magnesium. https://www.biolectra.com/magnesium-deficiency/consequences/eye-twitching/#:%7E:te xt=Magnesium%20relaxes%20muscles.,in%20other%20words%2C%20eye%20twitchin g.0
[4] Higuera, V. (2019, August 27). *What Causes Muscle Cramps?* Healthline. https://www.healthline.com/health/muscle-cramps#diagnosis
[5] Hill, R. A. D. (2019, November 21). *10 Interesting Types of Magnesium (and What to Use Each For)*. Healthline. https://www.healthline.com/nutrition/magnesium-types#10.-Magnesium-orotate
<6> Huebsch, T. (2020, June 26). *VIDEO: Frightening calf cramp appears to have a life of its own*. Canadian Running Magazine. https://runningmagazine.ca/sections/training/angel-bermudez-calf-cramp/0/
[7] Libretexts. (2020, August 22). *Chemistry of Magnesium (Z=12)*. Chemistry LibreTexts. https://chem.libretexts.org/Bookshelves/Inorganic_Chemistry/Supplemental_Modules_an d_Websites_(Inorganic_Chemistry)/Descriptive_Chemistry/Elements_Organized_by_Blo ck/1_s-Block_Elements/Group__2_Elements%3A_The_Alkaline_Earth_Metals/Z012_C hemistry_of_Magnesium_(Z12)
[8] *Magnesium*. (2021, October 14). The Nutrition Source. https://www.hsph.harvard.edu/nutritionsource/magnesium/
[9] *Magnesium deficiency*. (2021, May). Symptoms, Causes, Treatment & Prevention | Healthdirect. https://www.healthdirect.gov.au/magnesium-deficiency
[10] *Office of Dietary Supplements - Magnesium*. (2021, March 22). National Institutes of Health. https://ods.od.nih.gov/factsheets/Magnesium-Consumer/#:%7E:text=Magnesium%20is%20a%20nutrient%20that,protein%2C%20bone%2C%20and%20DNA
[11] *Osteoporosis: Calcium and Magnesium*. (2019, May 11). Spineuniverse. https://www.spineuniverse.com/conditions/osteoporosis/osteoporosis-calcium-magnesiu m
[12] Raman, M. R. S. (2018, June 9). *What Does Magnesium Do for Your Body?* Healthline. https://www.healthline.com/nutrition/what-does-magnesium-do#muscle-function
[13] S. (2019, December 30). *Magnesium and Bone Health*. Ask The Scientists. https://askthescientists.com/magnesium-bone-health/
<14> Suid-Kaap Forum News. (2016, October 19). *Suid-Kaap Forum News*. Suid-Kaap Forum. https://www.suidkaapforum.com/News/Article/LifeStyle/love-your-bones-20170711



6. The Role of Aerodynamics in Formula 1 Racing

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An Introduction to the Topic

Formula One or F1 is the highest class of international racing for formula racing cars. The racing cars used in this championship have been evolving since the first time F1 was inaugurated in 1950 by the FIA or Fédération Internationale de l'Automobile (Media, 2021). The cars utilize an aerodynamic-focused setup to stay competitive in the league. Aerodynamics is one of the integral factors in the performance of an F1 car which supports and maximizes each driver's ability to pilot highly accelerated vehicles around straights and tight corners. Due to the development of this factor, present F1 cars prove to have lower drag and greater downforce, which results in faster lap times as well as better stability when traveling at higher speeds. This is all in an effort to create tighter and better racing, which in turn provides a more exciting watching experience for motorsport fans worldwide. Aerodynamics are also necessary to ensure that the car moves as fast as possible despite large amounts of highspeed wind flowing in the opposite direction, which the car has to pass through.

Aerodynamics Principles Applied in F1 Cars

With any moving vehicle, such as an F1 car, there are forces present that act on the vehicle; the aerodynamic forces. There are four aerodynamic forces that act on all bodies moving through a fluid which are lift, weight (downforce in F1 cars), thrust, and drag. Lift is the force that acts on a body moving in a fluid opposite to the force of its weight or gravitational pull. In an F1 car, lift is an undesirable force as it reduces downforce and the grip of the tires. Weight is the effect of the Earth's gravitational pull on a moving body. F1 cars produce a certain type of force called downforce which pulls in the same direction as weight, the various components of an F1 car and its setup are designed with the goal of producing the maximum downforce possible while maintaining minimum drag. Team engineers and aerodynamicists are constantly trying to find ways in which downforce can be increased without drag (Pandit & Day, 2021, pp. 3-4).

Thrust is the force that allows the body to move through the fluid, it is exerted in the direction of where the body is moving towards. The greater the thrust, the greater the velocity of the body moving. The thrust comes from the raw power of the engine and the gearbox. Drag is the total opposite of thrust. It opposes the forces of thrust which means it acts as the friction of aerodynamics. The higher the drag, the slower the body will move. Drag is produced in two ways: through aerodynamic drag and induced drag (Fig. 1). Aerodynamic drag is the friction between the layers of the fluid and the surface in contact with the fluid due to the thickness of the fluid. For example, a thicker fluid like honey would produce more drag than a less thick fluid like water. Induced drag is caused by the pressure difference between the two ends of the body due to the fact that force acts from an area that consists of high pressure and moves towards an area of low pressure (Pandit & Day, 2021, p. 3).

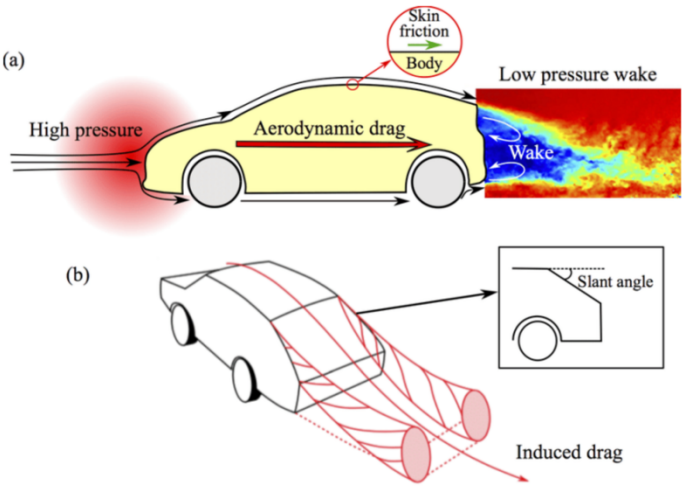


Figure 1. Sketch illustrating the drag origin for road vehicles: (a) aerodynamic drag, skin friction; (b) induced drag (Li, 2017).

How the 2022 F1 Regulations Utilize Aerodynamics

The main problem of racing in F1 previously was the loss of downforce. Based on research, previous F1 machines lose 35% of their downforce when racing behind another car approximately 20 meters behind (3 car lengths) and 47% of their downforce when racing approximately 10 meters behind (1 car length). This is the reason that cars aren't able to race closer. Now, the 2022 car is able to reduce those numbers to 4% at 20 meters behind and 18% at 10 meters behind (Stuart, 2021). The 2022 F1 regulations are the biggest overhaul in the sport for years, even though it was postponed for a year, it has now been introduced in 2022 to support closer racing which in turn creates more opportunities for cars to overtake each other. The regulations introduce changes to the front and rear wings and the floor of the car to create more downforce and grip (Cooper, 2022). The 2022 car also includes over-wheel winglets and wheel covers (a feature that was removed in 2009 but is back now). Wheel covers were included back again to create more airflow in the wheels for the cars to increase their downforce. The over-wheel winglets were added to direct the wake coming out of the front tires away from the rear wing. This role is usually done by the front wing but since that makes it sensitive when a car is racing behind another car, the over-wheelw winglets are here to substitute the front wing's job (Stuart, 2021).

The 2022 F1 car introduced a revamped front wing design. The job of the new front wing is to generate a consistent level of downforce when running closely behind another car. The front wing ensures that the wake of air coming from the front wheels is well controlled and directed down the car in the least disruptive way, steering the air narrowly down the side of the car as much as possible. There is no more gap present between the nose and the inboard end of the front wing, this change means that air will flow through the wing in a way that keeps it directed within the surface area of the car. Although the full-width elements of the front wing create a larger total wing surface area than before, it produces less downforce. The less downforce produced by the wing, the less it will disrupt the air and create turbulence, therefore the less sensitive the car will be to the disrupted airflow from the car in front (Stuart, 2021). Ground effect is a series of effects that have been exploited to create more downforce. Ground effect was prominent in the 1970s when cars were designed in the shape of upside-down aircraft wings to generate more downforce. In the

new 2022 car, they added a feature that would increase the use of ground effect which are the fully shaped underfloor tunnels that help generate larger amounts of downforce through ground effect (Stuart, 2021).

The vital aerodynamic component of an F1 car's rear assembly is the rear wing. The rear wing generates a great amount of downforce for the car through the air pressure difference its airfoils and shape create, however, it simultaneously generates drag that is turbulent for the car. Downforce is vital to F1 cars as it produces traction, giving the driver a better grip and increasing the velocity of the car. The position and shape of the 2022 F1 car's rear wing generate a rotational airflow that complies with the rear wheel wake and directs it into the flow exiting the diffuser, forming an invisible "mushroom"-shaped wake. This narrower wake is then projected high up into the air thanks to the airfoil flaps, allowing a pursuing car to drive through less disrupted clean air (Stuart, 2021).

To help visualize this information, Figure 2 shows the new 2022 car compared to last year.



Figure 2. A top-down comparison of the 2021 and 2022 F1 cars (Reynolds, 2021).

What Ferrari is Doing Right in 2022

The start of the F1 2022 season introduced many big modifications and changes to the rules and regulations of the sport, especially for cars. After the first few races of the season, it is quite clear that Ferrari has had the upper hand compared to the other teams on the grid, producing better race results and car performance. Entering the 8th race of the season, they stand second in the constructors' championship with only 36 points behind Red Bull. During the 2021 season, Ferrari shifted most of its resources and development towards the 2022 car for the next season since they weren't fighting for the championship (it was Red Bull and Mercedes last year). With that, they were able to create a car that would help them rise back to their glory. A key factor to the success of Ferrari this season is mainly their engine/power unit, the Ferrari 066/7. The engine delivers great performance such as high speeds and powerful torque without sacrificing reliability. Another key factor to Ferrari's success is the chassis design of their 2022 car. The Ferrari F1-75 has the best aerodynamic stability on the track compared to any of the other cars. It achieves this by avoiding using undercuts to direct air over the diffuser and preventing the outwash from the front from coming back inwards. A sharper nose cone also helps in reducing the amount of surface area in which the air makes contact (Ono, 2022). Fig. 3 shows the new Ferrari car in comparison with the 2021 car.

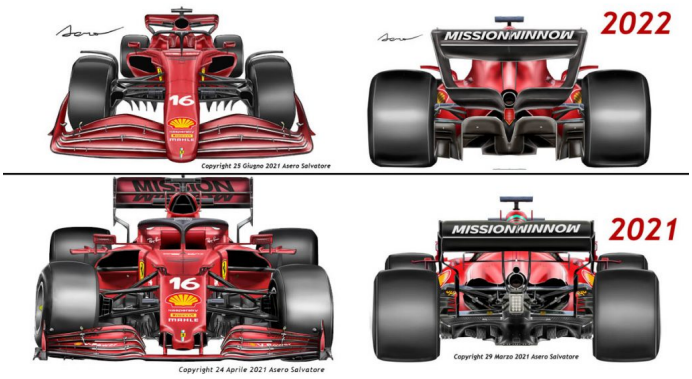


Figure 3. A Comparison of the front and rear of the 2021 and 2022 Ferrari F1 cars (Scuderia fans, 2021).

Conclusion

Through the years of racing in Formula One, the cars have developed and evolved to further support more exciting races for both drivers and fans. This evolution shows that F1 cars aren't merely racing cars that are fast but they include various factors of science like aerodynamics, and each factor needs to be maximized in order to create better cars. The sport is bigger than just the teams and the drivers, but also the engineers and aerodynamicists who work on the cars. This big change in the 2022 F1 car is just one of the steps in the sport's evolution, it is to be awaited that bigger and more revolutionary upgrades will arrive in the, hopefully, near future.

Glossary

Airfoil: a structure with curved surfaces designed to give the most favorable ratio of lift to drag in flight, used as the basic form of the wings, fins, and horizontal stabilizer of most aircraft.

Grid: a marked section of the track at the start where the cars line up according to their times in practice, the fastest occupying the front position.

Overtake: catch up with and pass while traveling in the same direction.

Traction: the grip of a tire on a road or a wheel on a rail.:

Wake: a trail of disturbed water or air left by the passage of a ship or aircraft.

References

Header Image: Assaf, R. (2021). A Formula 1 Car on a Race Track. https://www.pexels.com/photo/a-formula-1-car-on-a-race-track-10807493/

Cooper, S. (2022, April 1). From the safety car to bigger wheels, F1 rule changes for 2022 explained. iNews. Retrieved June 11, 2022, from https://inews.co.uk/sport/formula-one/f1-rules-2022-safety-car-bigger-wheels-formula-one-regulation-changes-explained-1519838

Li, R. (2017). [Sketch illustrating the drag origin for road vehicles: (a) aerodynamic drag, skin friction; (b) induced drag]. Research Gate. https://www.researchgate.net/figure/Sketch-illustrating-the-drag-origin-for-road-vehicles-a-aerodynamic-drag-skin_fig2_322640200

Ono, A. (2022, February 18). Analysis: The stand-out technical features on Ferrari's bold new F1-75 · RaceFans. RaceFans. Retrieved June 11, 2022, from https://www.racefans.net/2022/02/18/analysis-the-stand-out-technical-features-on-ferraris-bold-new-f1-75/Pandit, A., & Day, G. H. G. (2021). An Analysis and Survey on the Aerodynamics of F1 Car Design. Journal of Student Research, 10(2), 21. https://doi.org/10.47611/jsrhs.v10i2.1475

Reynolds, J. (2021, July 15). ANALYSIS: Comparing the key differences between the 2021 and 2022 F1 car designs | Formula 1®. F1. Retrieved June 11, 2022, from https://www.formula1.com/en/latest/article.analysis-comparing-the-key-differences-between-the-2021-and-2022-f1-car.4xYDht0jDee4cEQ3P4RsK9.html

Scuderia Fans. (2021). [A Comparison of the front and rear of the 2021 and 2022 Ferrari F1 cars]. Scuderia Fans. https://scuderiafans.com/f1-the-changes-and-upgrades-brought-by-ferrari-in-sochi-are-fundamental-for-2022/

Stuart, G. (2021, July 15). 10 things you need to know about the all-new 2022 F1 car | Formula 1®. F1. Retrieved June 11, 2022, from https://www.formula1.com/en/latest/article.10-things-you-need-to-know-about-the-all-new-2022-f1-car.4OLg8DrXyzHzdoGrbqp6ye.html

What is Formula 1? | What Is F1? | Formula 1 Racing. (2021, April 21). F1 Chronicle. Retrieved June 11, 2022, from https://f1chronicle.com/what-is-formula-1/



7. Maritime Monitoring via Low-Earth Orbit Satellite Constellation

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Abstract

With the ever-increasing volume of marine ships involved in commercial activities in an already globalised world, ensuring the efficiency and safety of these operations has become paramount. To accurately monitor their movement, satellite constellations are often used, making it crucial to optimise the performance of these constellations. However, there has been a lack of research done on the theoretical results of using large scale satellite constellations to monitor maritime activities. This report mainly investigates how key elements of a satellite constellation, such as the number of satellites per plane, number of planes, inclination, and combination, affect the coverage of the system. Experimentation and configuration will be done through Freeflyer software simulations. We hypothesise that increasing the inclination should increase the total number of maritime vessels observed but cause a decrease in the contact time of the satellite constellation with the ground stations. There should be a general increase in the contact time and the time taken to observe most of the satellites as the number of satellites, be it the number of planes or the satellites per plane, increases. More satellites per plane would also be more effective than more planes.

1.0 Methodology

1.1 Satellite constellation

For our simulations, the eccentricity and argument of perigee were assumed to be zero, and with reference to existing constellations, the altitude of satellites was 780km, and the cone half-angle was 17°. The satellites and their planes were equally distributed.

The inclination, number of satellites per plane, and the number of planes of the satellite constellation were varied. The inclination ranged from 70° to 85°, while the number of planes in each simulation ranged from 5 to 11, and the number of satellites per plane ranged from 5 to 15.

1.2 Ground Stations

To ensure continuous contact with ground stations and to mimic global demand for such a satellite constellation, ground stations were placed at the busiest container ports and at a diverse range of latitudes and longitudes.

1.3 Contact Analysis

Maritime vessels were simulated by point groups. Due to constraints, only 2064 maritime vessels were used, with a total simulation time of one day as the majority of marine vessels would have been detected.

The average time taken to observe 99% of our point groups and the average contact time of the satellite constellation with the ground stations were recorded to assess the performance of the constellation.

2.0 Results & Discussion

2.1 Effects of Inclination

Firstly, the impacts of changing inclination were investigated.

From Fig.1, the average contact time of the satellite constellation with the ground stations generally decreases with the increase of inclination, likely because the constellation would spend more time orbiting at higher latitudes, where fewer ground stations are.

Fig.2 shows that the number of ships detected by the constellation generally increases with inclination.

However, there is an anomaly at 75° of inclination, likely due to the difference in the initial position of the satellites which may

have caused the constellation to "miss" some of the point groups.

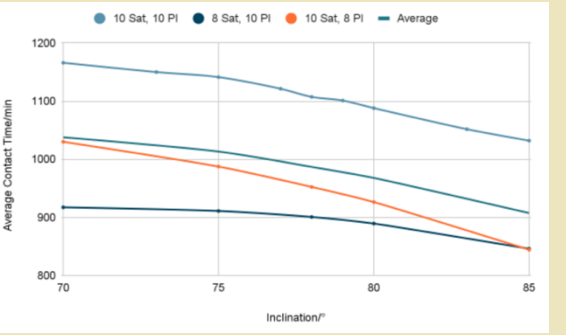


Figure 1. Graph of Average Contact Time Versus Inclination.

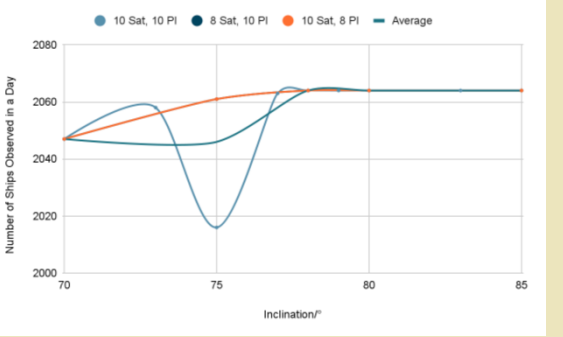


Figure 2. Graph of the Total Number of Ships Observed against Inclination.

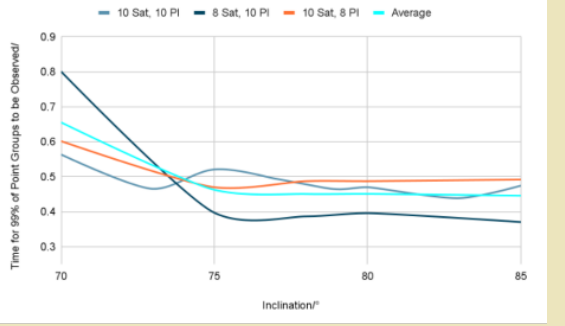


Figure 3. Graph of Time for 99% of Point Groups to be Observed against Inclination.

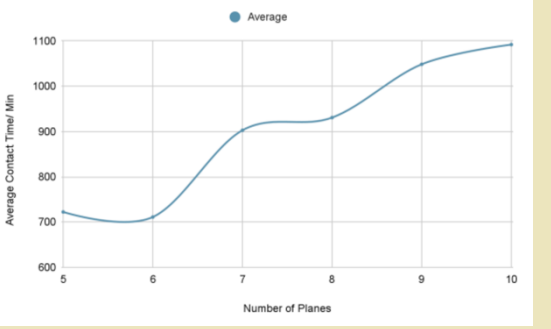


Figure 4. Graph of Average of Average Contact time against Number of Planes.

Next, the effect of inclination on the time taken to observe the majority of the ships was investigated.

Fig.3 suggests a negative relationship between inclination and the time taken to observe 99% of point groups because as inclination increases, the ground track of the satellites covers more latitudes, so more ships at higher latitudes are observed.

2.2 Effects of Change in Number of Satellite Planes

The inclination was then fixed at 80° and the effect of the number of planes on the constellation performance was investigated.

From Fig.4, the average contact time increased with the number of planes, though increasing from an odd number of planes to an even number had a negligible effect, falling partly within our expectations.

Fig.5 shows how the effect of the number of planes seems erratic, but a minimum time is observed at 8 and 10 planes.

The observation from Fig.5 is further indicated by Fig.6, which shows the apparent lack of pattern.

2.3 Effects of Change in Number of Satellite per Plane

A similar analysis of changes in the number of planes was done for changes in the number of satellites per plane.

The general increasing trend observed in Fig.7 is to be expected. However, from 9 satellites per plane to 10 satellites per plane, the average contact time only increased marginally.

This effect is seen more clearly in Fig.8, where a drop in the average contact time when the number of satellites per plane increases from 9 to 10 and the number of planes is even, not odd, can be observed. We suspect that this is due to different initial conditions.

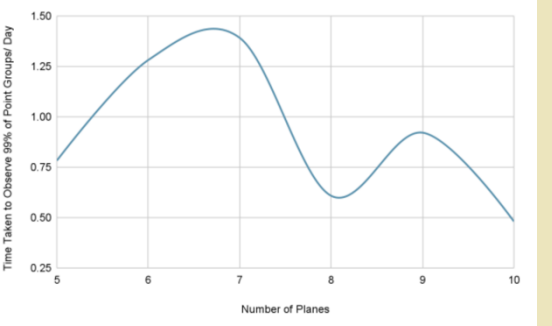


Figure 5. Graph of time taken to observe 99% of Point Groups against Number of Planes.

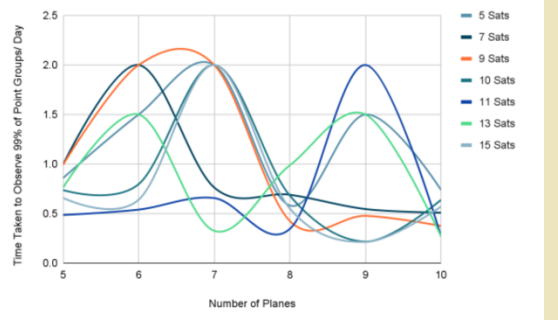


Figure 6. Graph of Time Taken for a Specific Number of Satellites per Plane to Observe 99% of Point Groups against Number of Planes.

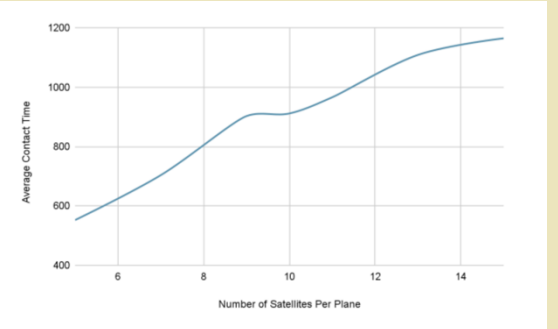


Figure 7. Graph of Average of Average Contact Time against Number of Satellites per Plane.

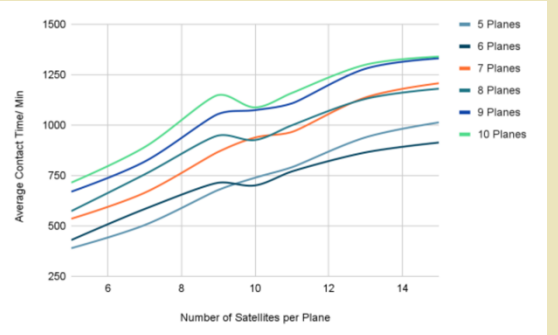


Figure 8. Graph of Average Contact Time against Number of Satellites per Plane

From Fig.9, it appears that a minimum average time taken to observe the majority of the point groups occurs at 7 and 11 satellites per plane, while a maximum point occurs at 9 and 13 satellites per plane.

2.4 Analysing the Effects of Number of Total Satellites

When determining the effects of changing the number of planes and the number of satellites per plane, questions on whether to increase the number of planes or the number of satellites per plane arise. We will use a table to discuss this.

With reference to Fig.10, for most combinations, having more satellites per plane is better than more planes, as stated in our hypothesis. However, an anomaly occurred in 7 orbital planes, likely due to the constellation configuration that results in more satellites being above land in the 24 hours of simulation.

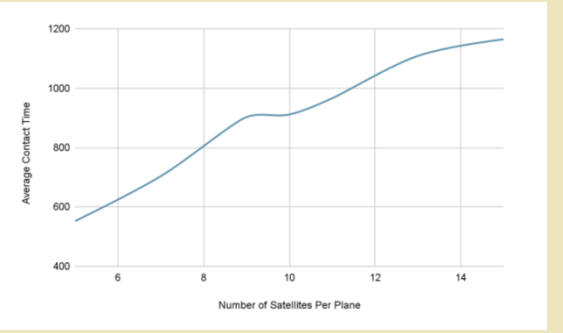


Figure 9. Graph of Time Taken to Observe 99% of Point Groups against Number of Satellites per Plane.

Figure 10. Table of Min Time Taken to Observe 99% of Point Groups/Day with Accordance to Number of Planes and Number of Satellites.

no. of planes / no. of sats	5	7	9	10
5		1.50	1.50	0.74
7	0.99		0.55	0.51
9	0.99	2.00		0.38
10	0.74	1.50	0.22	

3.0 Conclusion

From our previous discussions, the most efficient configuration of our satellite constellation has a 78° inclination, 10 orbital planes, and 11 satellites per plane. A larger number of satellites per plane is more beneficial.

3.1 Limitations

The increment for the increase in inclination and the step size of the simulation can be reduced to provide more accurate results.

Additionally, the combinations and range of inclination, the number of satellites per plane, and the number of orbital planes are non-exhaustive.

Costs were also not considered, resulting in a bias towards larger numbers of satellites, which may not be cost-effective.

References

Header Image: Mashkov, I. (2020). Radio telescope under bright starry sky. https://www.pexels.com/photo/radio-telescope-under-bright-starry-sky-6325003/